

Strong diffeomorphic approximation in three dimensions for $1 \leq p \leq 2$

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Abstract

We resolve the three-dimensional Ball–Evans approximation problem for $1 \leq p \leq 2$. Every $W^{1,p}$ homeomorphism between arbitrary bounded domains in $\mathbb{R}^3$ is a strong $W^{1,p}$ limit of smooth diffeomorphisms onto the same target.

1 Introduction and conventions

The Ball–Evans approximation problem concerns the compatibility of Sobolev approximation with global injectivity. A deformation in nonlinear elasticity is represented by a map between spatial domains; injectivity expresses noninterpenetration, whereas its weak first derivative records the local deformation. Smooth approximation is elementary if injectivity is discarded, but convolution need not preserve it. The question is whether the two requirements can be met simultaneously. Ball’s formulation explicitly attributes the question to Evans [4, Section 3, p. 8]; his later discussion and Hencl’s survey explain the surrounding variational questions [5, 22].

Here a *domain* is a nonempty connected open set. For domains $\Omega, \Lambda \subset \mathbb{R}^3$ and $1 \leq p < \infty$, a Sobolev homeomorphism is a homeomorphism $f : \Omega \rightarrow \Lambda$ whose components and weak first derivatives belong to $L^p(\Omega)$. A smooth diffeomorphism has a smooth inverse as well as a smooth forward map. The fixed-target version of the approximation problem asks for diffeomorphisms $f_j : \Omega \rightarrow \Lambda$ such that both $f_j - f$ and $Df_j - Df$ tend to zero in $L^p(\Omega)$. In particular, every approximating map, not just the limit, must have image exactly Λ .

We resolve this problem for the low and critical exponents in dimension three.

Theorem 1.1. *Let $\Omega, \Lambda \subset \mathbb{R}^3$ be nonempty bounded connected open sets, let $1 \leq p \leq 2$, and let $f : \Omega \rightarrow \Lambda$ be a homeomorphism in $W^{1,p}(\Omega; \mathbb{R}^3)$. There exist C^∞ diffeomorphisms $f_j : \Omega \rightarrow \Lambda$, each belonging to $W^{1,p}(\Omega; \mathbb{R}^3)$, such that*

$$\int_{\Omega} (|f_j - f|^p + |Df_j - Df|^p) dx \longrightarrow 0. \quad (1.1)$$

Both the map and its inverse in the conclusion are smooth on their open domains. No smoothness at the boundary, boundary extension of f , or regularity of either boundary is assumed. Neither inverse Sobolev regularity nor a nonvanishing Jacobian is an additional hypothesis. The target Λ is fixed throughout.

1.1 Prior work and the three-dimensional difficulty

In the plane, Iwaniec, Kovalev, and Onninen proved strong diffeomorphic approximation for $1 < p < \infty$ [26, Theorem 1.1]. Hencl and Pratelli established the endpoint $p = 1$, including

approximation by locally finite piecewise affine homeomorphisms [24, Theorem 1.1]. These theorems preserve the original target and do not require Sobolev regularity of the inverse. They show that the global topological constraint can be compatible with strong derivative control throughout the planar exponent range. The endpoint is a distinct achievement: strict convexity of the L^p norm is unavailable at $p = 1$.

The planar development also distinguishes two mechanisms. The $p = 2$ precursor of Iwaniec, Kovalev, and Onninen used harmonic replacements and smoothing [27]; their general $p > 1$ result used coordinatewise p -harmonic replacements. Hencl and Pratelli’s endpoint proof instead used geometric extension estimates and an adapted grid. Quantitative approximation of bi-Lipschitz planar maps by Daneri and Pratelli [15], and approximation of planar bi-Sobolev $W^{1,1}$ maps by Pratelli [36], additionally control inverse derivatives under their respective stronger hypotheses. Smooth invertibility of each approximant in Theorem 1.1 is not a claim of Sobolev convergence of the inverses.

Higher dimensions exhibit a genuine obstruction. Hencl and Vejnar initiated the counterexamples at $p = 1$ [25]. Campbell, Hencl, and Tengvall corrected a gap in that construction and extended the range to $1 \leq p < \lfloor n/2 \rfloor$ in dimensions $n \geq 4$, with Jacobian determinants of both signs on sets of positive measure [12]. A sequence of diffeomorphisms cannot reproduce this sign pattern by strong convergence of its derivatives. In dimension three, Hencl and Malý’s orientation theorem instead gives $\det Df \geq 0$ almost everywhere for a sense-preserving $W_{\text{loc}}^{1,1}$ homeomorphism [23, Theorem 1.1]. This sign conclusion does not assert a positive Jacobian and does not furnish an approximation: rank-deficient derivatives still require separate constructions.

Smoothing a homeomorphism that is already piecewise affine is an important intermediate problem. Mora-Corral and Pratelli obtained fixed-image diffeomorphic smoothing in the plane, with control of the forward map and its inverse [32]. Campbell and Soudský proved C^1 homeomorphic approximation of piecewise affine maps in arbitrary dimension [11]; their conclusion does not require the inverse to be smooth. Campbell, D’Onofrio, and Víttek subsequently proved diffeomorphic smoothing of locally finite piecewise affine homeomorphisms in dimensions three and four, including forward and inverse derivative-error control [13, Theorem A]. These results identify the importance of distinguishing a smooth injective map from a diffeomorphism. They do not construct a strong piecewise affine approximation of an arbitrary Sobolev homeomorphism.

The present proof separates that approximation problem from smoothing. It first constructs an onto locally bi-Lipschitz map with small strong Sobolev error, and then proves a smoothing theorem with arbitrarily fine pointwise control. Three-dimensional PL topology, developed by Moise and Bing and used here in Hamilton’s relative formulation, provides qualitative replacements [7, 19, 30, 31]. It supplies no derivative estimate. The energy bounds must therefore come from the parametrizations and measurements used below, rather than from the mere existence of a topological replacement.

The critical exponent $p = 2$ has additional analytic structure. Csörnyei, Hencl, and Malý proved that a three-dimensional $W_{\text{loc}}^{1,2}$ homeomorphism has an inverse of locally bounded variation, and that almost every parallel plane has the two-dimensional Lusin property [14, Theorems 1.1 and 1.3]. We use these two statements, respectively, to select reverse rays and to measure generic image surfaces. Neither is an assumption on the given inverse. The further analytic ingredients include Whitney’s extension theorem for compatible first jets [41] and conformal parametrization of metric disks [1, 29]. Their roles and the accompanying local estimates are made explicit at the point of use.

The complementary range $2 < p < \infty$ is treated by a separate construction in [34, Theorem 1.1]. The present paper does not invoke that range theorem or any of its proof modules. The smoothing theorem proved here holds for all finite exponents; the companion also includes a complete proof in its Appendix A.

1.2 Measurements, local models, and the proof strategy

The first objective is to replace f by a locally bi-Lipschitz homeomorphism while estimating derivatives without a quantitative topological extension theorem. We measure the images of a locally finite collection of source curves, and also of source surfaces when $p = 2$. A target triangulation gives complementary-dimensional dual pieces inside its tetrahedra, defined by ties among their largest barycentric coordinates. Intersections with these pieces determine a *budget*: each curve crossing is charged the length of a corresponding target edge, and each surface crossing is charged the area of a corresponding target face, with multiplicity. Generic sampling bounds these sums by the original trace integrals. Local topological changes make the counted crossings regular; a face-preserving target reparametrization then concentrates the image length or area near them. Thus Proposition 3.15 converts intersection counts into actual measurements for an onto locally bi-Lipschitz replacement. All protected local data are retained in buffered regions that are excluded from the measurements.

Measurements alone give an upper energy bound, not proximity to Df . We therefore retain finitely many disjoint compact sets on which the values and first derivatives of f are controlled and the derivative has a fixed positive rank. The discarded derivative integral is small; rank-zero points carry no such energy. The construction on these compact sets depends on the rank.

1. At full rank, compatible first jets extend to local C^1 diffeomorphisms. Radial source changes enlarge an exact germ, meaning agreement with such a model near the block center, into most of a small block, so that the map recovers the prescribed jet there. The surrounding thin region is filled using the trace budgets. At $p = 2$, reverse-ray selection and angular attenuation control the surface cost of this radial construction (Lemma 4.3).
2. At rank one or two, source coordinates split into the nonzero directions and the kernel of a nearby fixed matrix. Compressing the kernel reduces the controlled part of a volume block to a line or plane section. Nearly minimal curve length gives the correct derivative on a line. At rank two and $p = 2$, the sharp boundary-parametrized disk estimate gives the analogous conclusion from nearly minimal area (Lemma 3.3). At rank two and $p < 2$, a further radial construction inside the plane section provides the needed approximation.
3. Uncontrolled cells are reparametrized relative to their already chosen boundary values. Below a cell's dimension, radial collapse bounds its energy by the boundary energy (Lemma 3.4). This fills faces when $p < 2$ and volumes throughout $1 \leq p \leq 2$; the critical faces instead use the disk estimate. The resulting maps preserve the old cell images and agree on shared faces (Lemma 4.6).

The order of choices is part of the argument. Compact jet bounds and all finite multiplicative constants are fixed before the excess neighborhoods and trace errors are made small against them. The critical radial construction separates the regions whose area can be suppressed after replacement from those whose area must be measured in advance; Figure 2 displays these regions at the point where their construction is explained. Outside the retained compact configurations, universal mesh constants multiply only the small original energy tail. Summable local errors then yield a global $W^{1,p}$ approximation, not just a locally Sobolev map. For $p = 1$, the line argument uses an elementary squared length-defect estimate followed by Hölder's inequality, rather than strict convexity of L^1 .

Finally, locally bi-Lipschitz maps are approximated strongly by PL maps using finite local model libraries. Their derivative bounds are fixed before reducing the exceptional mesh volume. PL smoothing then uses small edge and vertex neighborhoods with summable costs. A fine value tolerance makes the smooth map properly homotopic to the original homeomorphism inside the

target; its positive local degrees and degree one force bijectivity. This proves the fixed image for each individual approximant, independently of passage to the limit.

Section 2 proves this smoothing theorem first. Section 3 develops the disk and collapse estimates, relative topological replacements, and sampled trace budgets. Section 4 constructs the rank-sensitive blocks and proves their global strong approximation estimate. Section 5 concludes the diffeomorphic approximation theorem on the original target.

1.3 Conventions

Unless stated otherwise, all domains, closures, local finiteness, and compactness are taken relative to the indicated open manifold. A compact set in an open domain has positive distance from its Euclidean complement. A PL map or triangulation is locally finite. A locally bi-Lipschitz homeomorphism has finite Lipschitz constants in both directions on sufficiently small neighborhoods; no global bound is implicit.

The symbol $|A|$ denotes the Frobenius norm of a matrix, and $\|A\|_{\text{op}}$ its operator norm. We use the Frobenius norm in sharp gradient estimates. Equivalent norms give the same strong convergence in Theorem 1.1. The symbols $\mathcal{L}^m$ and $\mathcal{H}^m$ denote Lebesgue and Hausdorff measure. Integrals on parameterized traces include their stated multiplicities. The differential along an m -dimensional parameter space is denoted D_m when it must be distinguished from the full differential.

A *fine value tolerance* on a domain U is a positive continuous function on U . To approximate finely means to meet a prescribed such tolerance pointwise. Whenever inverse control is needed on a compact localization, it is included in the prescribed tests. Locally finite constructions permit tolerances tending to zero toward the ends of the open domains; no positive global lower bound is assumed.

Constants called *absolute* depend only on dimension and on fixed universal reference shapes. A constant C_p may also depend on the fixed exponent. Other dependencies are specified when the constant is introduced. A finite constant that depends on a chosen compact collection of jets or charts is fixed before an error is required to be small against it.

A homeomorphism between connected oriented domains has a constant topological orientation. If necessary, reflect the target, perform the construction for the orientation-preserving map, and undo the reflection. This convention imposes no extra hypothesis on the Sobolev differential.

Lemma 1.2 (Local tolerances). *Let $U \subset \mathbb{R}^n$ be open and let $a : U \rightarrow (0, \infty)$ be continuous. For every $L > 0$ there is a positive L -Lipschitz function d on U such that*

$$d(x) \leq \min\{a(x), 1, L \text{dist}(x, \mathbb{R}^n \setminus U)\}.$$

There are also positive smooth minorants satisfying prescribed positive continuous upper bounds on their values and first derivatives. One may additionally impose positive constant bounds on a locally finite family of compact regional tests. Finite collections of such requirements can be combined, and error budgets on a countable compact exhaustion can be made summable.

Proof. Extend $e(x) = \min\{a(x), 1, L \text{dist}(x, \mathbb{R}^n \setminus U)\}$ by zero outside U , and put

$$d(x) = \inf_{y \in \mathbb{R}^n} \{e(y) + L|x - y|\}.$$

This function is L -Lipschitz and is bounded above by e . It is positive at an interior point x : on a small closed ball about x , e has a positive minimum, and outside that ball the distance term has a positive lower bound.

For a smooth minorant take a smooth locally finite partition of unity $\{\chi_i\}$ with compact supports in U and put $b = \sum_i c_i \chi_i$, with positive constants c_i . Given positive continuous bounds v, w , choose

$$c_i \leq 2^{-i-2} \min \left\{ \inf_{\text{supp } \chi_i} v, \frac{\inf_{\text{supp } \chi_i} w}{1 + \|D\chi_i\|_\infty} \right\}.$$

Then $b > 0$, $b \leq v$, and $|Db| \leq w$. Replace a finite list of requirements by their minimum. For countably many regional tests use a locally finite compact cover with compact enlargements, refine the bounds on each intersecting support, and allocate total errors by a convergent positive series. Only finitely many regional requirements meet a given compact support. $\square$

We will also use the following elementary global observation. If $H : U \rightarrow V$ is a homeomorphism and a continuous map $G : U \rightarrow V$ satisfies

$$|G(x) - H(x)| < \frac{1}{2} \text{dist}(H(x), \mathbb{R}^n \setminus V),$$

then the straight homotopy stays in V . When V is bounded, it is a proper homotopy: the preimage of a compact target set is contained, uniformly in the homotopy parameter, in the H -preimage of a compact set a positive distance from ∂V . This observation will be used together with local injectivity and degree, not as a substitute for local injectivity.

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2 Qualitative topology and strong smoothing

All PL structures in this Section are the ordinary structures on open subsets of $\mathbb{R}^3$. Triangulations are locally finite in the open domain. In particular, neither a triangulation nor the local constants used below need have uniform behavior at the boundary.

The qualitative PL tools belong to the three-dimensional triangulation and approximation theory of Moise and Bing [7, 30, 31]; Hamilton's relative formulation [19] is the precise interface used below. Edwards–Kirby's deformation theorem supplies the supported ambient extensions [17]. We separate these topological existence statements from the derivative estimates: finite local model libraries and subsequently chosen small exceptional sets provide the latter.

Theorem 2.1 (Strong smoothing of locally bi-Lipschitz maps). *Let $\Omega, \Omega' \subset \mathbb{R}^3$ be bounded domains, let $1 \leq p < \infty$, and let $H : \Omega \rightarrow \Omega'$ be a locally bi-Lipschitz homeomorphism in $W^{1,p}(\Omega; \mathbb{R}^3)$. For every $\varepsilon > 0$ and every continuous function $\xi : \Omega \rightarrow (0, \infty)$, there is a C^∞ diffeomorphism $g : \Omega \rightarrow \Omega'$ such that*

$$\|g - H\|_{W^{1,p}(\Omega)} < \varepsilon, \quad |g(x) - H(x)| < \xi(x) \quad (x \in \Omega). \quad (2.1)$$

In particular, g belongs to $W^{1,p}(\Omega; \mathbb{R}^3)$, and the approximations have exactly the target Ω' .

The proof has two parts. First we smooth a locally finite PL homeomorphism with summable local derivative errors. We then approximate H by such a PL map. In the second part a finite collection of local models gives derivative bounds before the measure of the exceptional mesh cubes is made small. The qualitative topology used to choose those models supplies no derivative estimate.

2.1 Relative qualitative tools

Lemma 2.2 (Fine relative PL approximation). *Let M, N be PL three-manifolds without boundary, with N metrized, and let $f : M \rightarrow N$ be a homeomorphism. Suppose that $K \subset M$ is closed and that f is PL on an open neighborhood of K . Given a continuous function $\eta : M \rightarrow (0, \infty)$, there is a PL homeomorphism $f_1 : M \rightarrow N$ which agrees with f on a neighborhood of K and satisfies*

$$d_N(f_1(x), f(x)) < \eta(x) \quad (x \in M).$$

Noncompact manifolds are allowed. For manifolds with boundary the same statement holds if K contains ∂M and f is PL near K .

Proof. Choose a closed neighborhood K^+ of K contained in the open set on which f is PL. Transport the PL structure of N to M by f . The identity from the original structure to the transported structure is PL near K^+ . Hamilton's relative approximation Theorem [19, Section 3, Theorem 2.2, pp. 68–69] applies to these two structures. Use the compatible metric $d_f(x, y) = d_N(f(x), f(y))$ and the fine tolerance η . It gives a homeomorphism a that is PL between the two structures, is the identity on K^+ , and satisfies $d_f(a(x), x) < \eta(x)$. Then $f_1 = f \circ a$ has all the asserted properties. The boundary condition in Hamilton's Theorem is precisely the additional condition stated here; for an open Euclidean domain its manifold boundary is empty. $\square$

Lemma 2.3 (Small local ambient extension). *Fix a closed Euclidean cube $B \subset \mathbb{R}^3$, compact sets $C, D \subset B$ with $C \subset \text{int } B$, and an open neighborhood O of $C \cup D \cup \partial B$. For every $\lambda > 0$ there is a*

$\delta > 0$, depending only on these source sets and on λ , with the following property. If $e : O \rightarrow \mathbb{R}^3$ is an embedding,

$$\sup_{x \in O} |e(x) - x| < \delta, \quad e = \text{id} \text{ on } D \cup \partial B,$$

then there is a homeomorphism $A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ such that

$$A = e \text{ on } C, \quad A = \text{id} \text{ on } D \cup (\mathbb{R}^3 \setminus \text{int } B), \quad \sup_{\mathbb{R}^3} |A - \text{id}| < \lambda. \quad (2.2)$$

The number δ is uniform over a finite list of fixed normalized configurations (B, C, D, O) . No derivative bound on e is required.

Proof. Put $E = C \cup D \cup \partial B$. The deformation Theorem of Edwards–Kirby [17, Theorem 5.1], applied at the inclusion of O in $\mathbb{R}^3$, deforms every sufficiently close embedding e through embeddings e_t , starting at $e_0 = e$, so that $e_1 = \text{id}$ on E . The deformation is unchanged outside a fixed compact neighborhood L of E in O , and fixes the inclusion. Its continuity at the inclusion permits us to require

$$|e_t(x) - x| < \lambda/2 \quad (x \in L, 0 \leq t \leq 1), \quad |e(x) - x| < \lambda/2 \quad (x \in L).$$

A sufficiently small uniform δ on O meets the finitely many compact-open conditions that specify this neighborhood of the inclusion. In the terminology of that Theorem a proper embedding respects manifold boundaries; this condition is automatic for the ambient manifold $\mathbb{R}^3$.

All the open images $e_t(O)$ coincide. To see this, enclose L in a finite union of relatively compact subdomains of O whose boundary collars are outside the region where the deformation changes the map. The embeddings agree on that collar. Invariance of domain [21, Theorem 2B.3] and degree relative to the common boundary show that the images of the enclosed subdomain are the same. Outside it the maps already agree. On this common open image define

$$A_0 = e \circ e_1^{-1},$$

and define A_0 to be the identity elsewhere. The map is already the identity off the compact set $e_1(L)$ in the common image, so this is an ambient homeomorphism. It agrees with e on E and has displacement less than λ . Since it fixes ∂B pointwise, it preserves the bounded complementary component $\text{int } B$. Restrict A_0 to B and extend it by the identity outside B to obtain A .

The construction uses only the displayed source configuration and continuity at the inclusion. For finitely many normalized configurations take the minimum of the resulting positive tolerances. $\square$

We will use Lemma 2.3 when the input embedding is defined by a small map η near a new core and by the identity near older protected sets. The following observation specifies the required gluing condition. Choose open sets $\overline{O_0} \subset U_0$, where η is defined on U_0 , and an open set V containing the protected sets, such that $\eta = \text{id}$ on $U_0 \cap V$. If η is sufficiently close to the identity that $\eta(O_0) \subset U_0$, the map

$$e = \eta \text{ on } O_0, \quad e = \text{id} \text{ on } V \quad (2.3)$$

is an embedding. Indeed, a collision $\eta(x) = y$ with $x \in O_0$ and $y \in V$ has $y \in U_0 \cap V$, whence $\eta(y) = y$ and injectivity of η gives $x = y$. The open embedding property then follows from invariance of domain. All the sets in this observation can be fixed in advance when the source configurations range over a finite list.

Lemma 2.4 (Fine control, properness, and the target). *Let $h : \Omega \rightarrow \Omega'$ be an orientation-preserving homeomorphism between bounded domains, and let $g : \Omega \rightarrow \mathbb{R}^3$ be a smooth map with $\det Dg > 0$. If*

$$|g(x) - h(x)| < \frac{1}{2} \text{dist}(h(x), \mathbb{R}^3 \setminus \Omega') \quad (x \in \Omega), \quad (2.4)$$

then g is a diffeomorphism of Ω onto Ω' .

Proof. Write $d(y) = \text{dist}(y, \mathbb{R}^3 \setminus \Omega')$ and $g_t = (1-t)h + tg$. Condition (2.4) puts the entire segment $g_t(x)$ in Ω' . Since d is 1-Lipschitz,

$$d(g_t(x)) \leq \frac{3}{2}d(h(x)).$$

If $K \subseteq \Omega'$ and $m = \min_K d > 0$, the inverse image of K under the homotopy lies in

$$h^{-1}(\{y \in \Omega' : d(y) \geq 2m/3\}) \times [0, 1].$$

The set in braces is compact because Ω' is bounded. Thus g_t is a proper homotopy. Its endpoint g has the same degree as h , namely one. A proper local diffeomorphism has finitely many preimages of each point, and here each preimage has positive local degree. The degree-one identity therefore says that every point of Ω' has exactly one preimage. The local smooth inverses patch to a smooth inverse on Ω' . $\square$

2.2 Smoothing a locally finite PL homeomorphism

Proposition 2.5 (Strong PL smoothing). *The conclusion of Theorem 2.1 holds if H is replaced by a locally finite PL homeomorphism $h : \Omega \rightarrow \Omega'$ in $W^{1,p}(\Omega; \mathbb{R}^3)$.*

Proof. An orientation-reversing Euclidean isometry in the target reduces the proof to the orientation-preserving case. Fix a locally finite affine triangulation for h . We prescribe summable error budgets for its edges, vertices, and a locally finite family of compact sets covering the remainder of the domain. All modifications below can also obey an arbitrary positive continuous value tolerance.

The open edges. Around each open edge choose a tube of radius $d(t)$, where $0 < t < l$ is its axial coordinate. The tubes of distinct open edges are disjoint, their radii are bounded by a small multiple of $\min(t, l-t)$, and d is exactly linear near both endpoints. These choices follow from the finite geometry of each simplex star; local finiteness makes them compatible throughout Ω . We may make $\|d'\|_\infty$ small by decreasing the width.

In positively oriented orthogonal frames, and after translations, the original map on this tube is

$$(t, r, \theta) \mapsto (at + rb(\theta), rs(\theta)e_{\phi(\theta)}), \quad e_\alpha = (\cos \alpha, \sin \alpha), \quad (2.5)$$

where $a > 0$, $s > 0$, and the coefficients are smooth on the closed angular sectors. The transverse homogeneous map is injective: otherwise two equal transverse images, taken at arbitrarily small radius, would have their axial difference canceled by changing t , contradicting injectivity of h in the edge star. Its angular map is consequently an orientation-preserving circle homeomorphism. We choose a lift ϕ with $\phi(\theta + 2\pi) = \phi(\theta) + 2\pi$. Positive determinants on the finitely many sectors give upper and positive lower bounds for s and for the one-sided derivatives ϕ' .

Choose constants $c > 0$ and c' , and for $0 \leq \tau \leq 1$ put

$$b_\tau = (1-\tau)b, \quad s_\tau = (1-\tau)s + \tau c, \quad \phi_\tau = (1-\tau)\phi + \tau(\theta + c').$$

For fixed τ , the determinant of the corresponding map in the frame $(\partial_t, \partial_r, r^{-1}\partial_\theta)$ is

$$as_\tau^2 \phi'_\tau > 0. \quad (2.6)$$

On all the closed sectors and all $\tau \in [0, 1]$ this has a positive minimum. Let χ be a smooth cutoff equal to 1 on $(-\infty, -1]$ and to 0 on $[-1/2, \infty)$, and substitute

$$\tau(t, r) = \chi\left(\frac{\log(r/d(t))}{N}\right)$$

in the preceding formula. The additional derivative columns have size at most

$$\frac{C}{N}(1 + \|d'\|_\infty). \quad (2.7)$$

Indeed, $|r\partial_r\tau| \leq C/N$ and $|r\partial_t\tau| \leq C(r/d)|d'|/N$, while $r/d \leq 1$ on the support of the modification. The remaining factors come from the fixed angular data. Taking N large preserves positive determinants on every sector, including one-sided limits. Near the axis the result is the nonsingular affine map $(t, r, \theta) \mapsto (at, cre_{\theta+c'})$. Near the tube boundary it agrees with h .

All first derivatives are bounded by an edge-dependent constant independent of a further common decrease of d . The volume of the tube tends to zero under that decrease. Both its derivative error and its value error therefore have arbitrarily small $W^{1,p}$ cost. Choose these costs summably over the edges and call the result h_1 . It is continuous and locally Lipschitz, fixes all vertices, and has the required small total error. Because the tube widths are exactly linear near endpoints, $h_1 - h_1(v)$ is homogeneous of degree one on a sufficiently small ball about every vertex v .

The faces and the punctured vertex balls. Away from the vertices, the closed convex hull of the nearby almost-everywhere gradients of h_1 is contained in $GL^+(3)$ on a sufficiently small neighborhood of each point. At a smooth point this follows by continuity. At a nonsmooth point the only remaining interface is one face. If A_- and A_+ are its two gradient traces, continuity gives agreement on the tangent plane, so $A_+ - A_-$ has rank at most one with normal covector to that face. Consequently

$$\det((1-t)A_- + tA_+) = (1-t)\det A_- + t\det A_+ > 0 \quad (0 \leq t \leq 1). \quad (2.8)$$

The traces vary continuously on the two sides. Their nearby convex hull lies in a sufficiently small neighborhood of this compact positive-determinant segment, proving the assertion. Homogeneity makes this property uniform at relative scale on a punctured ball about a vertex: cover the unit sphere by finitely many of the corresponding neighborhoods and rescale.

Let $\rho \geq 0$ be a smooth probability kernel supported in the unit ball. Away from vertices define

$$h_2(x) = \int_{\mathbb{R}^3} h_1(x + w(x)z)\rho(z) dz, \quad (2.9)$$

where w is smooth and positive there and the integration balls lie in Ω . Its derivative is

$$Dh_2(x) = \int Dh_1(x + w(x)z)\rho(z) dz + \int (Dh_1(x + w(x)z)z) \otimes Dw(x)\rho(z) dz. \quad (2.10)$$

The first integral is in the positive-determinant convex hull just described. Taking w and $|Dw|$ sufficiently small locally makes the second integral a small perturbation, and hence gives $\det Dh_2 > 0$. Formula (2.9) is smooth where $w > 0$, as is also seen by writing its smooth variable kernel in the integration variable $y = x + w(x)z$.

On a small punctured ball at v choose $w(x) = c_v|x - v|$, with $c_v > 0$ sufficiently small. This is consistent with the relative-scale convex-hull condition and makes $h_2 - h_1(v)$ homogeneous on a smaller ball. Set $h_2(v) = h_1(v)$ there. The required radius function exists by a locally finite

partition of unity with sufficiently small positive coefficients. Near the vertices blend this function with $c_v|x - v|$ on disjoint annuli. The annular cutoff terms are bounded by a constant times c_v if the competing radii are chosen at that same small scale, so both the radius and gradient restrictions survive. This is also an instance of the local minorant construction in Lemma 1.2.

Here is the global error choice. First take disjoint vertex balls so small that their volumes, multiplied by the fixed local gradient bounds to the power p , meet a summable error budget. On these balls the gradients in Equation (2.10) are bounded independently of a further decrease of their radii. On a compact set outside the balls, the local gradients are bounded and converge at every piecewise smooth point as $w, |Dw| \rightarrow 0$. Dominated convergence therefore gives an arbitrarily small derivative error on that compact set. A locally finite compact covering with summable budgets makes the total $W^{1,p}$ error as small as prescribed. The value error follows from local Lipschitz continuity and the small radius. These arguments include $p = 1$.

The vertices. Center a homogeneous vertex ball at the origin and subtract $h_1(v)$. Write the resulting map as F . Euler's identity is $DF(x)x = F(x)$. Since $DF(x)$ is invertible for $x \neq 0$, $F(x)$ never vanishes there. We can thus write

$$F(r\theta) = rR(\theta)\Phi(\theta), \quad R > 0, \quad \theta \in S^2. \quad (2.11)$$

Its positive determinant implies that $\Phi : S^2 \rightarrow S^2$ is a smooth orientation-preserving local diffeomorphism. It is a covering by compactness, and it is one-sheeted because S^2 is simply connected.

Smale's Theorem gives a smooth isotopy Φ_τ from Φ to a rotation, smooth jointly in (τ, θ) [38, Theorem 6]; for the jointly smooth deformation see also [28, Theorem 1.5]. Interpolate R through positive functions R_τ to a positive constant. For fixed τ the homogeneous map $F_\tau(r\theta) = rR_\tau(\theta)\Phi_\tau(\theta)$ has determinant

$$R_\tau(\theta)^3 J_{S^2} \Phi_\tau(\theta) > 0.$$

The family has bounded first derivatives and a positive determinant minimum, by compactness. On an arbitrarily small ball replace F by $F_{\tau(r)}$, where τ is 0 near the outer boundary, 1 near the center, and $\sup_r |r\tau'(r)|$ is sufficiently small. Such a transition is obtained by spreading a fixed cutoff over a sufficiently long interval of $\log r$. Its extra derivative term is bounded by $C|r\tau'(r)|$, so positive determinant persists. At the center the map is a dilation followed by a rotation. It is therefore smooth and nonsingular there.

The derivative bound for this last modification depends on the fixed spherical isotopy but not on the support radius. Shrinking that radius gives arbitrarily small summable $W^{1,p}$ costs over all vertices. The resulting map g is smooth and has positive determinant throughout Ω .

Finally, in each of the three operations impose local value tolerances whose sum is less than

$$\min \left\{ \xi(x), \frac{1}{2} \text{dist}(h(x), \mathbb{R}^3 \setminus \Omega') \right\}.$$

The preceding constructions permit these fine tolerances and a total Sobolev error less than ε . Lemma 2.4 makes g a diffeomorphism onto the original target. This proves the Proposition. $\square$

Remark 2.6. Theorem A of Campbell–D'Onofrio–Vítek [13] also gives diffeomorphic approximation of locally finite piecewise affine homeomorphisms in dimension three. Proposition 2.5 records the specific local and fine-control construction needed here; the reduction from a general locally bi-Lipschitz map is proved next.

2.3 Fine meshes and controlled patch insertion

It remains to approximate H strongly by a PL homeomorphism. Near most points, an affine first-order model will control the new derivative. On the remaining patches, qualitative topology provides replacements but gives no bound for their slopes. Choosing those replacements only after making the exceptional patches small would therefore leave the energy estimate circular. We instead select finite libraries of replacements on each coarse neighborhood, fixing their derivative bounds before the final mesh resolution. The mesh and insertion lemmas below let neighboring replacements retain exactly the same data on their overlaps.

Fix a coarse, locally finite Whitney decomposition $\mathcal{P}$ of Ω into closed dyadic cubes with disjoint interiors. Choose relatively compact enlarged neighborhoods U_P that are still locally finite. Enlarge them a fixed finite number of times when necessary. Since H is locally bi-Lipschitz, there are constants $K_P \geq 1$ such that

$$K_P^{-1}|x - y| \leq |H(x) - H(y)| \leq K_P|x - y| \quad (x, y \in U_P). \quad (2.12)$$

Enlarge K_P also to bound $|DH|$ almost everywhere on U_P in the chosen matrix norm. The constants may incorporate the data from finitely many neighboring coarse cubes. All references below to data local to P allow this finite enlargement, but never an enlargement depending on the final fine-mesh resolution.

Lemma 2.7 (Adapted dyadic meshes). *There is an absolute integer n_0 with the following property. Given arbitrary positive local upper bounds on the fine scale, there is a locally finite dyadic cube decomposition $\mathcal{Q}$ of Ω with centers c_Q and side lengths l_Q for which, on writing*

$$B_Q(s) = c_Q + [-sl_Q, sl_Q]^3, \quad (2.13)$$

the following hold:

- (i) $B_Q(100) \subseteq \Omega$, and cubes meeting $B_Q(50)$ have side lengths comparable to l_Q by absolute constants.
- (ii) The normalized cube configurations in $B_Q(6)$ belong to a finite list. There is a conforming affine triangulation of $\mathcal{Q}$ with finitely many normalized nondegenerate shapes on these configurations.
- (iii) The cubes have n_0 colors so that the closed boxes $B_Q(6)$ of a single color are pairwise disjoint.
- (iv) Attach to Q a coarse index $P(Q)$ containing c_Q , using a fixed convention on coarse boundaries. The initial scale bounds can be reduced so that all interactions through any fixed number of neighboring patch layers are confined to a fixed locally finite graph of coarse indices, independent of further mesh refinement. The corresponding patches lie in the neighborhoods on which the required bounds in Equation (2.12) hold.

The upper bounds on the fine scale remain arbitrarily prescribable.

Proof. Choose a positive size function s on Ω with sufficiently small absolute Lipschitz constant, below all the prescribed local bounds and below $10^{-3} \text{dist}(x, \mathbb{R}^3 \setminus \Omega)$. Lemma 1.2 supplies such a minorant. Take the maximal dyadic cubes for which

$$l_Q \leq \inf_Q s.$$

Failure of the condition for the dyadic parent gives $s(x) \leq (2 + C \text{Lip}(s))l_Q$ on Q . The small Lipschitz constant then gives local comparability on $B_Q(50)$. For example, choosing it sufficiently

small makes every dyadic ratio in this neighborhood belong to $\{1/2, 1, 2\}$. The distance bound puts $B_Q(100)$ inside Ω . Maximality gives a decomposition, and the positive lower bound for s near every compact subset gives local finiteness.

The cube vertices have dyadic coordinates. Bounded local scale ratios and bounded normalized distance therefore give only finitely many normalized configurations. To triangulate, decompose each cube face into its square contact facets with adjacent cubes. Put every contact vertex and hanging subdivision point on the perimeter of each such square. Cone these perimeter segments from the square center, and then cone the resulting triangulation of each cube boundary from the cube center. The constructions on shared facets agree. They give nondegenerate simplices, and their normalized shapes range over a finite list.

Local comparability bounds the degree of the graph joining cubes whose $B_Q(6)$ boxes meet. A coloring with one more than that degree proves (iii). Finally choose the initial local upper bounds so small relative to the coarse Whitney sizes that the finitely many required patch layers stay in fixed coarse neighborhoods. Since the coarse enlarged cover is locally finite, its interaction graph is locally finite. Taking a still smaller minorant s does not change any of these conclusions. $\square$

The finite list in Lemma 2.7 records actual normalized geometric configurations, including the positions of their faces and vertices. When protected boxes are subsequently shrunk by one of finitely many fixed amounts, their intersections and all the margins used below still range over finitely many geometric configurations.

Lemma 2.8 (Insertion with protected cores). *Consider a patch Q and a finite collection of older protected boxes with the relative sizes and positions provided by Lemma 2.7. For every older box prescribe a larger half-shrunk box and a smaller fully-shrunk box, separated by a fixed positive normalized margin. Let $G : \Omega \rightarrow \Omega'$ be a homeomorphism, and let h_Q be a PL embedding on a neighborhood of $B_Q(3)$. Assume that $h_Q = G$ on the intersection of $B_Q(3)$ with each half-shrunk box.*

For every $\lambda > 0$ there is an $a > 0$, depending only on λ , the normalized source configuration, and a local constant K_P in Equation (2.12), such that

$$\sup_{B_Q(5)} |G - H| < al_Q, \quad \sup_{B_Q(3)} |h_Q - H| < al_Q \quad (2.14)$$

implies the following conclusion. There is a domain homeomorphism A supported in $B_Q(5)$, with

$$\sup |A - \text{id}| < \lambda l_Q,$$

such that $G \circ A = h_Q$ on a neighborhood of $B_Q(2)$ and $G \circ A = G$ on the fully-shrunk older boxes. The tolerances are uniform over a finite list of source configurations. They do not depend on the derivatives of G or of h_Q .

Proof. Scale the patch to $l_Q = 1$. First reduce a using K_P so that $h_Q(B_Q(3)) \subset G(\text{int } B_Q(4))$. Indeed, for $x \in B_Q(3)$ and $z \in \partial B_Q(4)$, the distance $|H(z) - H(x)|$ is bounded below by K_P^{-1} . The small errors in Equation (2.14) preserve this separation. Degree on $\text{int } B_Q(4)$, comparing G with H on the boundary, shows that $h_Q(x)$ is in its image. The same statement holds with a small margin around $B_Q(3)$.

The embedding

$$\eta = G^{-1} \circ h_Q$$

is consequently defined there. If $z = \eta(x)$, then

$$|z - x| \leq K_P |H(z) - H(x)| \leq K_P (|H(z) - G(z)| + |h_Q(x) - H(x)|) < 2K_P a. \quad (2.15)$$

Thus its closeness to the inclusion uses no inverse-Lipschitz bound for G .

Take a fixed closed neighborhood C of $B_Q(2)$ and fixed open neighborhoods O_0, U_0 with $C \subset O_0 \Subset U_0 \Subset \text{int } B_Q(3)$, and let D be the union of the fully-shrunk older boxes intersected with $B_Q(5)$. The half/full shrink margins give neighborhoods on which the map η on U_0 agrees with the identity near D . Include also an identity neighborhood of $\partial B_Q(5)$, separated from C . Choose these neighborhoods once for each normalized source configuration. By Equation (2.15), sufficiently small a makes the union of η and these identity maps an embedding, as in Equation (2.3). Apply Lemma 2.3 with $B = B_Q(5)$ and displacement λ . Precomposition by the resulting A installs h_Q on C and preserves the fully-shrunk boxes. Rescaling proves the Lemma. $\square$

We record explicitly how tolerances in repeated insertions are chosen. For a patch insertion supported in $B_Q(5)$,

$$|G(A(x)) - H(x)| \leq |G(A(x)) - H(A(x))| + K_P |A(x) - x|. \quad (2.16)$$

After division by the local cube size, interacting patches introduce only the bounded ratios from Lemma 2.7. Hence, for a fixed number of stages, all tolerances can be assigned by backward recursion: start with the desired final value bounds; choose a small allowed displacement at the last stage; use Lemma 2.8 to obtain its input tolerance; require the preceding value error to be a small fraction of that tolerance; and continue backward. At a coarse index take the minimum over its finitely many interacting indices at each step. Only finitely many steps occur. Every tolerance remains positive, and none depends on a PL derivative bound selected later.

2.4 Finite PL libraries before resolution selection

Finite normalized PL model families with exact agreement on earlier overlaps appear in Väisälä's implementation of Carleson's method [39, Sections 1.2 and 2.7–2.11]. Here they are combined with the relative insertion just proved and a subsequent choice of mesh resolution to obtain strong derivative control.

Proposition 2.9 (Strong PL approximation of a locally bi-Lipschitz map). *Under the hypotheses of Theorem 2.1, there is a locally finite PL homeomorphism $G : \Omega \rightarrow \Omega'$ satisfying*

$$\|G - H\|_{W^{1,p}(\Omega)} < \varepsilon, \quad |G(x) - H(x)| < \xi(x) \quad (x \in \Omega).$$

Proof. Fix the coarse neighborhoods, local bounds K_P , and finite mesh configurations of Lemma 2.7. Until the final resolution choice, we describe rules and libraries uniformly over all admissible meshes, rather than fix a particular mesh. Write $S = 2n_0$ for the number of stages. A newly installed patch has protected radius 2. At each subsequent stage decrease the radius of every older protected patch by

$$d_* = (4n_0 + 4)^{-1} \quad (2.17)$$

in its own B_Q coordinates. Its final radius is greater than $3/2$, and in particular its closed original cube Q remains in the interior of its protected core. At an insertion we use half of the current prescribed decrease for agreement of the new model and the old map, and the full decrease for the sets fixed by the ambient change. These are precisely the margins in Lemma 2.8.

Choose all insertion and displacement tolerances by the backward procedure following Equation (2.16). They can be chosen to give an arbitrarily small final normalized value error on every patch. If $a_{s,P}$ is the accuracy required of a new model at stage s , require the current $G - H$ bound before that stage to be less than $a_{s,P}/8$, also on its interacting neighborhoods. This additional fraction is imposed during the same backward recursion. All these numbers are now fixed.

Quantized affine interpolation and good patches. Choose positive thresholds δ_P so small that errors $C\delta_P$ suffice at every first-round insertion involving that coarse neighborhood, and so that

$$\sum_{P \in \mathcal{P}} (C\delta_P)^p |U_P| < \varepsilon^p/8. \quad (2.18)$$

Here and below a local tolerance is reduced using finitely many neighboring indices when necessary. We also require $C\delta_P < (2K_P)^{-1}$, where C is the interpolation constant for the finite mesh shapes.

At each vertex v of the fine triangulation, round $H(v)$ to a target dyadic grid whose spacing is a small dyadic multiple of the smallest incident cube size. The dyadic factor is fixed from the coarse indices of those cubes, small enough that all the corresponding normalized rounding errors are at most δ_P . Interpolate the rounded values affinely on the conforming triangulation to obtain a continuous piecewise affine map L . We do not assert that L is globally injective.

Call a cube Q , with $P = P(Q)$, good if there is an affine map A_Q of bi-Lipschitz constant at most $2K_P$ such that

$$\sup_{B_Q(20)} |H - A_Q| \leq \delta_P l_Q, \quad \left(\frac{1}{|Q|} \int_Q |DH - DA_Q|^p \right)^{1/p} \leq \delta_P. \quad (2.19)$$

Changing the fixed factor 20 by a larger absolute factor would have the same effect. The rounding and the finite shape list imply

$$\begin{aligned} |DL - DA_Q| &\leq C\delta_P \quad \text{near } B_Q(3), \\ \sup_{B_Q(3)} |L - H| &\leq C\delta_P l_Q. \end{aligned} \quad (2.20)$$

For completeness, subtract A_Q at the vertices of any such tetrahedron. The vertex errors are at most $C\delta_P l_Q$; the inverse matrix of its three edge vectors has norm at most C/l_Q , by finite normalized shapes. Multiplication gives the first estimate. The value estimate follows by affine interpolation and Equation (2.19).

The first estimate in Equation (2.20) makes L an embedding on a neighborhood of $B_Q(3)$. Indeed, $L - A_Q$ is Lipschitz on a slightly larger convex box with constant at most $C\delta_P$; integrate its piecewise constant derivatives on segments. Thus

$$|L(x) - L(y)| \geq ((2K_P)^{-1} - C\delta_P)|x - y| > 0.$$

Invariance of domain gives the open embedding assertion.

The first round. Start with $G_0 = H$. For stages $1, \dots, n_0$ process the colors in order, inserting a patch only if it is good, and taking its model to be $h_Q = L$. On every overlap with an older protected core this agrees with the already installed map, because both equal the single global map L . Equation (2.20) and the chosen thresholds supply the input accuracy for Lemma 2.8. Supports $B_Q(5)$ of one color are disjoint, so these insertions are simultaneous. Their local finiteness makes the union a domain homeomorphism. The value invariants follow from Equation (2.16). At the end of this round the map agrees with L on every good cube and on a protected neighborhood of it.

Normalized data and their finiteness. For each Q choose $m_Q \in \mathbb{Z}^3$ nearest to $H(c_Q)/l_Q$ and use normalized coordinates

$$z = \frac{x - c_Q}{l_Q}, \quad \hat{H}_Q(z) = \frac{H(c_Q + l_Q z) - l_Q m_Q}{l_Q}. \quad (2.21)$$

The value at $z = 0$ is bounded by an absolute constant, and Equation (2.12) gives a uniform local Lipschitz bound. The normalized vertex values of L on $B_Q(4)$ lie in a bounded set on finitely many dyadic grids. They therefore take only finitely many values for each coarse index. Here it matters that the translation $l_Q m_Q$ is included in the data: without it one would not have finiteness of affine maps, as opposed to finiteness of their gradients.

If R interacts with Q , then

$$\frac{l_R m_R - l_Q m_Q}{l_Q} \quad (2.22)$$

is a bounded dyadic vector with a denominator from a finite list. Boundedness follows by comparing $H(c_R)$ and $H(c_Q)$ using Equation (2.12); the denominators are fixed by the bounded dyadic scale ratios. Consequently the first-round exact PL data, as viewed in any interacting normalized patch, have only finitely many possibilities.

Choosing the second-round libraries. For stages $n_0 + 1, \dots, 2n_0$ we insert at every patch of the current color. The models are chosen as follows, before the final fine scale is selected. For each coarse index, the normalized maps $\widehat{H}_Q$ on a fixed larger patch form a uniformly bounded equicontinuous family. They admit finitely many sup-norm bins of any prescribed positive diameter. At stage s use diameter less than $a_{s,P}/8$.

Inductively assume that the already installed normalized PL data have finite lists. The relative positions of the protected cores have finitely many possibilities, by Lemma 2.7 and Equation (2.17). Their target translations have finitely many possibilities by Equation (2.22). Thus the exact older data on all required half-shrunk overlaps have a finite list at this stage as well. There are consequently only finitely many combinations of

$$\text{source configuration, normalized } H\text{-bin, and exact protected PL data.} \quad (2.23)$$

For each combination which can occur, select one representative normalized embedding G^* , close to a map H^* in that bin by the prescribed current value tolerance, and realizing the specified older data. Take the representative on the fixed open patch $(-4, 4)^3$. If $\widehat{K}_i$ are the normalized half-shrunk older boxes, the set

$$K = [-3, 3]^3 \cap \bigcup_i \widehat{K}_i$$

is compact in that open patch. It lies strictly inside the currently protected cores, so G^* is PL on a neighborhood of K . Apply Lemma 2.2 to G^* on the open patch, keeping K fixed. Obtain a PL embedding h^* there with uniform error less than $a_{s,P}/8$ on the required compact patch. Store its restriction to a neighborhood of $[-3, 3]^3$ in the library.

If a different actual tuple has the same combination in Equation (2.23), this single model agrees with all of its prescribed PL overlap data. Moreover,

$$\|h^* - \widehat{H}_Q\|_\infty \leq \|h^* - G^*\|_\infty + \|G^* - H^*\|_\infty + \|H^* - \widehat{H}_Q\|_\infty < \frac{3}{8} a_{s,P}. \quad (2.24)$$

The three norms are on the fixed patch needed for insertion. The actual current map is already within $a_{s,P}/8$ of its H , so Lemma 2.8 applies after undoing the normalization. This installs h_Q while preserving the fully-shrunk older cores.

This selection does not require compactness of the family of possible intermediate maps G . Only the H -family is binned; the intermediate map is used as an existence witness for an embedding realizing the exact finite overlap data. All possible normalized meshes and inputs obeying the fixed coarse bounds may be included when deciding which combinations occur. At any stage the

choices for a coarse index depend only on libraries from strictly earlier stages at its finitely many interacting indices, so they can be made simultaneously over all coarse indices. Hence the choices are independent of the eventual mesh resolution.

Each stored map is PL on a neighborhood of a fixed compact patch and thus has finitely many affine pieces there. It has a finite upper derivative bound. There are finitely many stored maps for each local coarse combination, and only finitely many stages. This proves by induction both the finiteness of the exact data needed at the next stage and the existence of constants $M_P < \infty$ such that the final map satisfies

$$|DG| \leq M_P \quad \text{almost everywhere on } Q, \quad P = P(Q). \quad (2.25)$$

The constants M_P are fixed before reducing the fine scale. They may be arbitrarily large; no efficient dependence on K_P is asserted.

After the second round every cube has a protected PL neighborhood. The final map is locally PL. If a single triangulation is desired, take common refinements of the finitely many affine subdivisions meeting each compact set and triangulate the resulting polyhedral cells. Local finiteness gives a locally finite affine triangulation on Ω . The map is still onto Ω' because every stage was a precomposition by a domain homeomorphism. On every good cube the map still equals L , since its first-round protected neighborhood was preserved throughout.

Selecting the resolution and paying for bad cubes. Only now choose the final local upper size bounds of Lemma 2.7. On a fixed coarse compact region, almost every x is both a differentiability point of H and a Lebesgue point of DH . The derivative at such a point is invertible, with the upper and lower bounds inherited from local bi-Lipschitzness. The affine map

$$A_x(y) = H(x) + DH(x)(y - x)$$

then passes both tests in Equation (2.19) on every sufficiently small cube containing x . Indeed its enlargement is contained in a ball of radius Cl_Q about x , which gives the sup-norm test by differentiability. The volume of that ball is at most a fixed multiple of $|Q|$, which gives the derivative mean test by the Lebesgue-point property. Only finitely many coarse thresholds occur near a fixed compact set.

It follows that the measure of the union of bad cubes with a fixed coarse index P tends to zero as their local upper side length tends to zero. This statement is uniform over the admissible meshes. One can see the uniformity by taking the union of all bad dyadic cubes with that index and side at most r . These measurable unions decrease as $r \downarrow 0$, and their intersection is null by the preceding pointwise argument. They lie in the fixed finite-volume neighborhood U_P .

Choose positive numbers b_P with $\sum_P b_P < \varepsilon^p/8$, and reduce the local mesh bounds until

$$(M_P + K_P)^p \left| \bigcup_{\substack{Q \text{ bad} \\ P(Q)=P}} Q \right| < b_P. \quad (2.26)$$

All bounds can be imposed simultaneously using Lemma 2.7. The M_P in this inequality were fixed in Equation (2.25); they are unaffected by this refinement.

On good cubes, Equation (2.19) and Equation (2.20) give

$$\int_Q |DG - DH|^p \leq (C\delta_P)^p |Q|.$$

On bad cubes use Equation (2.25) and Equation (2.12). Summing and using Equation (2.18) and Equation (2.26) makes the derivative error less than the allocated part of ε^p .

The stage tolerances also bound $|G - H|/l_Q$ by fixed local constants on each cube. Further reduce the local size bounds so that the actual value error is below ξ and has p th integral less than the remaining part of ε^p . For example require it to be below a global constant $\varepsilon/[4(1 + |\Omega|)^{1/p}]$ and below the positive minimum of $\xi/2$ on the relevant compact neighborhood. These reductions only help the bad-cube estimate. The map is locally Lipschitz, its derivative is globally p -integrable by the estimates just proved, and its values lie in the bounded target. Thus it belongs to $W^{1,p}(\Omega; \mathbb{R}^3)$ and has the required strong error. $\square$

Proof of Theorem 2.1. Apply Proposition 2.9 with Sobolev tolerance $\varepsilon/2$ and value tolerance $\xi/2$, obtaining a PL homeomorphism $h : \Omega \rightarrow \Omega'$. Apply Proposition 2.5 to h with the same tolerances. The triangle inequality proves Equation (2.1). Both approximations have exactly the target Ω' . Choosing $\varepsilon \downarrow 0$ gives the sequence asserted by the smoothing reduction. $\square$

3 Parametrization, topology, and trace budgets for the low exponents

The constructions in this Section use length measurements when $1 \leq p < 2$, and both length and area measurements when $p = 2$. Area always means parametrized Euclidean area, counted with multiplicity. The Dirichlet integral uses the squared Frobenius norm, without a factor $1/2$. All topological changes are made in the interiors of the open domains. In particular, none of the relative statements below concerns an extension to the boundary of either domain.

The disk and collapse Lemmas convert compatible boundary data and image measurements into energy bounds for parametrizations. The topological Lemmas prepare crossings and exact germs; the simplex squeeze then converts the counted crossings into trace measurements. Source and target sampling control the resulting budgets.

3.1 Boundary parametrization of a disk

The conversion of area into nearly minimal Dirichlet energy by a change of parameters has a classical precursor in Morrey's ε -conformal parametrization theorem [33, Chapter I, Section 9, Theorem 1.2]. Here we also need exact boundary agreement and control of the boundary energy.

Definition 3.1 (Essential tangential compatibility). Let D be a compact polygonal disk or a convex polyhedral cell and let $v : D \rightarrow \mathbb{R}^d$ be Lipschitz. We say that v has essential tangential compatibility if the following properties hold.

1. At almost every interior point x , the a.e. derivative of v has essential limit $Dv(x)$ as its argument tends to x .
2. In radial coordinates based at an interior point, at almost every boundary parameter the derivatives in boundary directions have essential limits, from the interior as well, equal to the derivatives of the boundary restriction.

The exceptional sets here are null sets on the relevant parameter manifold. An ambient null-set assertion alone is insufficient for the second condition.

Lemma 3.2 (Compatibility of finite closed-piece formulas). *A continuous Lipschitz map given on a compact parameter manifold by finitely many C^1 formulas extending to neighborhoods of their*

closed pieces has the compatibility in Definition 3.1. The same assertion applies to compositions of such maps. It continues to hold after a boundary parametrization is changed on each edge by a bi-Lipschitz map whose derivative has an essential limit almost everywhere, and the change is extended by coning.

Proof. At a point belonging to a closed piece, only pieces containing that point can meet every sufficiently small neighborhood of it. At almost every point of an overlap, that point is a density point of the overlap in its parameter manifold. Two C^1 extensions whose values agree there have equal derivatives in its tangent directions: their difference vanishes on a set of density one, and its first-order expansion therefore vanishes on every tangent vector. Continuity of the finitely many extension derivatives now gives the asserted essential limits. Apply this argument on the boundary parameter manifold for boundary directions. For a composition, partition by the finitely many choices of formulas and use the chain rule; tangential agreement on a preimage overlap follows from the same density argument. In a cone extension, away from its center and face rays the formula is a smooth radial factor times the boundary formula. The asserted limits consequently follow from those of the boundary derivative; the excluded rays and parameter exceptions have measure zero. $\square$

Lemma 3.3 (Boundary-parametrized disk estimate). *Put $Q = [-1/2, 1/2]^2$. Let $h : Q \rightarrow \mathbb{R}^d$ be a bi-Lipschitz embedding having essential tangential compatibility, and put*

$$A(h) = \int_Q J_2 Dh, \quad L(h) = \int_{\partial Q} |D_t h|^2.$$

For every $\varepsilon > 0$ there is a bi-Lipschitz self-homeomorphism $\rho : Q \rightarrow Q$, equal to the identity on ∂Q , such that

$$\int_Q |D(h \circ \rho)|^2 \leq C(A(h) + L(h)) + \varepsilon, \quad (3.1)$$

where C is absolute.

There is also the following sharp form. Suppose h_j are such embeddings, $\sup_j L(h_j) < \infty$, $A(h_j) \leq 1 + o(1)$, and $h_j|_{\partial Q} \rightarrow \iota|_{\partial Q}$ uniformly, where $\iota(x) = (x, 0) \in \mathbb{R}^d$. The parametrizations can be chosen so that

$$\int_Q |D(h_j \circ \rho_j)|^2 \leq 2 + o(1), \quad D(h_j \circ \rho_j) \longrightarrow D\iota \quad \text{in } L^2(Q). \quad (3.2)$$

The bi-Lipschitz constants of the chosen parametrizations may depend on the individual embedding. The conclusions transport to polygons through fixed shape-controlled, bi-Lipschitz piecewise smooth charts; the sharp square normalization is used when the constant 2 matters.

Proof. We first majorize the pullback metric by a smooth metric that is conformally Euclidean on an added collar. Conformal coordinates on a rectangle then give the weak energy estimate. For the sharp estimate, coordinate tests force both the rectangle and its boundary correction to be nearly isometric.

Metric preparation. Fix $0 < \tau \leq 1$, set $S = 1 + 2\tau$ and $Q^+ = SQ$, and use square radial coordinates. For $\delta > 0$, let $r : Q^+ \rightarrow Q$ preserve radial directions, carry the inner Q by the homothety of factor $1 - \delta$, and carry the added shell linearly along each radius onto $Q \setminus (1 - \delta)Q$.

Thus r is bi-Lipschitz for fixed positive δ . Given $\xi > 0$, we may choose δ and a smooth positive metric g on a neighborhood of Q^+ so that

$$g \geq D(h \circ r)^T D(h \circ r) \quad \text{a.e.}, \quad (3.3)$$

$$g = M^2 I \quad \text{throughout } Q^+ \setminus Q, \quad (3.4)$$

$$\text{area}_g(Q) \leq A(h) + \xi, \quad \text{area}_g(Q^+ \setminus Q) \leq C\tau L(h) + \xi, \quad (3.5)$$

$$\|h \circ r - h\|_{C^0(\partial Q)} \leq \xi, \quad \int_{\partial Q} |D_t(h \circ r)|^2 \leq CL(h) + \xi. \quad (3.6)$$

Here a scalar factor in (3.4) is allowed to vary smoothly in the shell.

We give the details of this preparation because its boundary assertion is used at the critical exponent. On the shell the radial derivative of $h \circ r$ is bounded by $C\delta \text{Lip}(h)/\tau$. Essential tangential compatibility implies that the essential suprema of the perimeter-direction derivatives in shrinking neighborhoods of almost every boundary parameter tend to the corresponding boundary derivative norm. They are bounded by fixed Lipschitz data. Covering the perimeter by short overlapping intervals, taking suprema on slightly enlarged intervals, and using a smooth partition of unity therefore gives scalar majorants whose squared integrals are at most $CL(h) + \xi$ once δ is sufficiently small. The negligible radial derivative can be included in these majorants. Square radial and ordinary angular coordinates have uniformly bounded distortion in this shell. Their conformal metrics consequently have area at most $C\tau L(h) + \xi$. Choose in addition a typical sufficiently small inner level to obtain (3.6).

On the inner square consider the pullback quadratic form in (3.3). It is bounded, uniformly positive for the fixed data, and essentially continuous almost everywhere. On a fine partition of unity, majorize it by a representative matrix on each enlarged patch plus its essential oscillation times the identity and a small positive scalar matrix. The resulting smooth positive form dominates the original form. The oscillations tend to zero almost everywhere and are bounded, so its area integral tends to that of the pullback form by dominated convergence. The latter is the area of h on $(1 - \delta)Q$, hence is at most $A(h)$. Call this inner majorant g_0 , and call the scalar shell majorant $M^2 I$. Choose a finite number N dominating the operator norms of g_0 , $M^2 I$, and both one-sided pullback forms near ∂Q . This is possible for the fixed map and the fixed positive δ . In a two-sided neighborhood of ∂Q of thickness t set the metric equal to NI . Blend g_0 to NI on the inner side and NI to $M^2 I$ on the outer side by smooth cutoffs supported in a slightly larger neighborhood. Each blend is between two forms dominating the relevant pullback form, so it still dominates that form; the entire outer blend is scalar. The metric is constant on an open neighborhood of the interface, so its jets match across the sides and corners. Smooth neighborhoods and cutoffs can be chosen with area $O(t)$, and the added metric area is at most CNt . Choose $t < \tau/10$ and then t so small that CNt is below the remaining error allowance. Derivatives of the cutoff metric are not charged to any estimate. Along a sequence, N may diverge, but t is chosen after N so that $Nt \rightarrow 0$. In particular a fixed-width outer band of the added shell is unaffected. This proves (3.3)–(3.6).

Conformal coordinates. By isothermal coordinates and marked-quadrilateral uniformization [1, Lemma 7 and Theorem 6][29, Theorem 4.31], uniformize the smooth metric quadrilateral (Q^+, g) conformally onto the centered rectangle

$$R_{W,S} = [-W/2, W/2] \times [-S/2, S/2],$$

with corresponding corners, and write this map as Φ . The marked quadrilateral uniformization determines the modulus W/S ; a Euclidean dilation sets the height to S . In the interior Φ is smooth

and nonsingular. We justify this regularity before using Φ in the energy estimates. The metric's Beltrami coefficient μ is smooth and vanishes on the outer scalar band, so it extends by zero to a compactly supported smooth coefficient on the plane. Lemma 7 of Ahlfors–Bers gives a C^1 solution with positive Jacobian. Its smooth bootstrap follows from their parameter construction [1, Section 2, Theorem 2]: choose $q > 2$ with $\|\mu\|_\infty C_q < 1$ for the norm C_q of their singular integral operator. Translated coefficients $\mu_t(z) = \mu(z + t)$ vary smoothly in $L^\infty \cap L^q$; the uniformly invertible operator in that construction therefore gives smooth dependence of $U_t - \text{id}$ in its solution space B_q , where $U_t(0) = 0$ and $\partial U_t - 1 \in L^q$. The Hölder norm of B_q makes point evaluation continuous. Writing $U = U_0$, uniqueness gives $U_t(z) = U(z + t) - U(t)$. On any bounded neighborhood of t , choose a fixed z_* such that $z_* + t$ stays outside $\text{supp } \mu$. Then $U(t) = U(z_* + t) - U_t(z_*)$ is smooth, since its first term is holomorphic there and its second depends smoothly on t . The conformal rectangle map preserves interior smoothness and nonsingularity. Since the metric is conformally Euclidean near the outer boundary, Schwarz reflection across sides [2, Chapter 4, Section 6.5, Theorem 24], and then across the two perpendicular sides at each corner, shows that Φ and its inverse are bi-Lipschitz up to the boundary for each fixed metric.

Weak estimate. Fix, for example, $\tau = 1/4$. Extend the first and second Euclidean coordinate functions from the outer boundary through the shell using cutoffs which vanish before reaching Q . Their g -Dirichlet integrals are bounded by an absolute constant: on the shell the scalar factor cancels from a two-dimensional Dirichlet integral. On the rectangle, the corresponding horizontal and vertical boundary differences equal S . Cauchy's inequality on horizontal and vertical segments gives, respectively, the lower bounds

$$S^3/W \quad \text{and} \quad SW. \quad (3.7)$$

It follows that W is bounded above and below by positive absolute constants.

Choose $\eta < c\tau$, for example $\eta = \tau/16$. Because g is scalar on the whole added shell, Φ is ordinary holomorphic there. Reflect in each outer source side and its corresponding rectangle side. Near a corner use the two perpendicular reflections, which commute. Every point in the resulting neighborhood of width η folds into the original shell, so these formulas give one analytic extension on that entire neighborhood. Away from its seams the derivative is an original nonzero derivative or its conjugate. Across a side the reflected halves map to opposite half-planes, and at a corner the four copies map into the four distinct target quadrants. The extension is therefore locally injective also on seams and corners, and its derivative is nonzero throughout the band. Its image is contained in $[-3W/2, 3W/2] \times [-3S/2, 3S/2]$, since at most two side reflections are used. The rectangle bounds and Cauchy's estimate on disks in the band now give a uniform upper derivative bound on a slightly narrower band, independent of the metric on the inner square.

Choose C strictly above that upper bound. The positive harmonic function $u = \log(C/|\Phi'|)$ is defined on this fixed-width band. On each outer side the integral of $|\Phi'|$ equals the corresponding rectangle side length. Thus some boundary point has $|\Phi'| \geq c_0 > 0$. A fixed finite chain of disks along the outer boundary gives by Harnack's inequality $u \leq C_H \log(C/c_0)$ there. Consequently $|\Phi'| \geq C \exp(-C_H \log(C/c_0)) > 0$ on the entire boundary, including the corners. The constants depend on the fixed τ , but not on the inner metric distortion. The boundary map of Φ thus has a uniformly controlled bi-Lipschitz extension $P : Q^+ \rightarrow R_{W,S}$, obtained by coning from the centers.

Set, for $x \in Q$,

$$F(x) = h \circ r \circ \Phi^{-1} \circ P(Sx).$$

Since $P = \Phi$ on ∂Q^+ and $r(Sx) = x$ there, this is $h \circ \rho$ with ρ a bi-Lipschitz self-homeomorphism fixing ∂Q . The homothety does not change two-dimensional Dirichlet energy. Conformal invariance,

the bounded distortion of P , and (3.3) yield

$$\int_Q |DF|^2 \leq C \int_{Q^+} |d(h \circ r)|_g^2 d\text{area}_g \leq 2C \text{area}_g(Q^+).$$

Letting ξ be sufficiently small proves (3.1).

Sharp estimate. First fix an arbitrarily small positive τ and make the metric errors tend to zero along the given sequence. Denote the first two physical coordinates of $h_j \circ r_j$ on Q by X_j, Y_j . Each coordinate has squared g_j -gradient at most one, so each has energy at most $\text{area}_{g_j}(Q) \leq 1 + o(1)$. By (3.6), its boundary discrepancy from the corresponding Euclidean coordinate is bounded in $W^{1,2}(\partial Q)$ and tends to zero in $L^2(\partial Q)$. The interpolation inequality

$$\|v\|_{H^{1/2}(\partial Q)}^2 \leq C \|v\|_{L^2(\partial Q)} \|v\|_{H^1(\partial Q)}$$

therefore makes the discrepancy tend to zero in $H^{1/2}$. The trace extension operator on the fixed polygonal collar, followed by a cutoff, extends it with vanishing Euclidean $W^{1,2}$ norm and with support away from the outer half of the shell. Extend X_j, Y_j accordingly so that they equal the Euclidean coordinates on that outer half. Their individual shell energies are at most $S^2 - 1 + o(1)$, since the metric is scalar there. Their total individual energies on Q^+ are thus at most $S^2 + o(1)$.

Apply (3.7) to these tests on $R_{W_j, S}$. Both S^3/W_j and SW_j are at most $S^2 + o(1)$; hence $W_j \rightarrow S$. Moreover, if $(\hat{X}_j, \hat{Y}_j) = (X_j, Y_j) \circ \Phi_j^{-1}$, equality asymptotically in the two segmentwise inequalities gives

$$D(\hat{X}_j, \hat{Y}_j) \rightarrow I \quad \text{in } L^2(R_{W_j, S}). \quad (3.8)$$

For example, the horizontal integral of $\partial_1 \hat{X}_j$ on every segment is S , and subtracting S^3/W_j from its energy gives exactly

$$\int_{R_{W_j, S}} (|\partial_1 \hat{X}_j - S/W_j|^2 + |\partial_2 \hat{X}_j|^2).$$

The vertical coordinate has the analogous identity with derivative 1.

On the outer coordinate band, the pair in (3.8) equals Φ_j^{-1} . For an orientation-preserving similarity A in dimension two,

$$|A^{-1} - I|^2 \det A = |A - I|^2.$$

Change of variables therefore gives $D\Phi_j \rightarrow I$ in Euclidean L^2 on the corresponding source band. Reflection and the analytic interior estimates give uniform derivative convergence on a smaller band containing the boundary. The corner normalization and $W_j \rightarrow S$ also give convergence of the values there. Blend Φ_j in this fixed band with the linear map $(x_1, x_2) \mapsto (W_j x_1/S, x_2)$. This gives extensions P_j of the boundary values with distortion $1 + o(1)$; for large j their derivatives are close to the identity, so they are bi-Lipschitz homeomorphisms onto the rectangle. The preceding energy argument now gives

$$\int_Q |D(h_j \circ \rho_j)|^2 \leq (1 + o(1)) 2 \text{area}_{g_j}(Q^+) \leq 2 + C\tau \sup_j L(h_j) + o(1).$$

Take $\tau \downarrow 0$ diagonally. Finally, boundary agreement and uniform convergence of the boundary values give, by integration in coordinate directions,

$$\int_Q D(h_j \circ \rho_j) : D\iota \rightarrow 2.$$

Expanding the square of the derivative difference and using the energy bound proves the second conclusion in (3.2). $\square$

3.2 Subcritical collapse onto boundary data

The skeleton and radial-extension strategy is classical in Sobolev approximation; see Bethuel [6, Section II.1, pp. 164–165, and Section II.3, Eqs. (32), (37), pp. 170–171]. This background motivates the collapse below. Its injective reparametrization, fixed image and exact boundary compatibility are proved locally; Bethuel’s density theorem does not supply these homeomorphism requirements.

Lemma 3.4 (Cell collapse). *Let $m \in \{2, 3\}$ and $1 \leq p < m$. Let $D \subset \mathbb{R}^m$ be a convex polyhedral cell of scale ℓ , with inradius bounded below by $c\ell$ and diameter bounded above by $C\ell$. Suppose $H : D \rightarrow \mathbb{R}^d$ is a bi-Lipschitz embedding with essential tangential compatibility. Let $\sigma : \partial D \rightarrow \partial D$ be a face-preserving, cellwise bi-Lipschitz homeomorphism. For every $\varepsilon > 0$ there is a bi-Lipschitz parametrization $F : D \rightarrow H(D)$ with $F|_{\partial D} = H \circ \sigma$ and*

$$\int_D |DF|^p \leq C_{m,p,c,C} \ell \int_{\partial D} |D_t(H \circ \sigma)|^p + \varepsilon. \quad (3.9)$$

The boundary data on shared cells are retained exactly, so the parametrizations glue when those data agree.

Proof. Rescale to $\ell = 1$ and place an incenter at the origin. Cone-extend σ from that center; this is bi-Lipschitz for the fixed data. In radial coordinates $x = rz$, $z \in \partial D$ and $0 \leq r \leq 1$, precompose the resulting map by the radial homeomorphism with profile

$$q_{\alpha,\delta}(r) = \begin{cases} (1 - \delta)r/\alpha, & 0 \leq r \leq \alpha, \\ 1 - \delta + \delta(r - \alpha)/(1 - \alpha), & \alpha \leq r \leq 1. \end{cases}$$

Here $0 < \alpha, \delta < 1$, and the resulting parametrization is

$$F_{\alpha,\delta}(rz) = H(q_{\alpha,\delta}(r)\sigma(z)).$$

On $r \leq \alpha$ all derivatives are at most a fixed-data constant times α^{-1} ; hence the energy there is $O(\alpha^{m-p})$. On $r > \alpha$ the radial derivative tends to zero as $\delta \downarrow 0$, while the tangential derivatives tend essentially to $r^{-1}D_t(H \circ \sigma)(z)$. Indeed, the angular derivative contains $q_{\alpha,\delta}(r)DH(q_{\alpha,\delta}(r)\sigma(z))D_t\sigma(z)$. The last factor is evaluated at the fixed boundary point z ; only the base map’s tangential derivatives require a limiting argument. The radial Jacobian is comparable to r^{m-1} with constants determined by the cell shape. Essential compatibility and dominated convergence yield an upper bound

$$C \left(\int_{\alpha}^1 r^{m-1-p} dr \right) \int_{\partial D} |D_t(H \circ \sigma)|^p$$

for the limiting outer energy. To apply the essential limits, use a sequence of the individual bi-Lipschitz radial changes and the common full-measure set of their Lebesgue representatives. Coning σ only inserts its fixed a.e. angular derivative. Since $p < m$, the radial integral stays bounded as $\alpha \downarrow 0$ and the central error tends to zero. First choose α and then δ . Restoring scale multiplies the boundary integral by ℓ and proves (3.9). All radial profiles fix the boundary, which proves the gluing assertion. $\square$

3.3 Local topological modifications

The budget construction needs regular transverse crossings at its measured hits, while the radial blocks need exact identity germs at selected centers. Both are local topological requirements; neither

provides an energy estimate. We first treat one crossing. We then control the marked rays meeting at a center, where the way they join across a surrounding shell must also be retained.

All constructions in this subsection take place in interior coordinate neighborhoods of three-manifolds. The classical inputs are the locally flat Schoenflies theorem, Dehn's lemma, and the classification of mapping classes of punctured spheres; see [9, 10, 18, 35]. We use the relative approximation and small embedding extension statements of Lemmas 2.2 and 2.3. Marked points on a sphere carry no tangent framing; an isotopy fixing such a point need not fix a tangent direction there.

Lemma 3.5 (Cleaning an isolated crossing). *Let P be a locally flat embedded surface disk and let I be an interval which is straight in specified local topological coordinates. Suppose that $z \in P \cap I$ is interior to both objects, is an isolated point of their intersection, and has a neighborhood containing no other surface sheet under consideration. There is an ambient modification of P , supported in an arbitrarily small neighborhood of z , after which the intersection in that neighborhood is either empty or consists of a single standard transverse crossing. At a retained crossing the disk can be taken flat across the straight interval. No initial local flatness of the pair (P, I) is assumed.*

If the interval passes from one local side of P to the other at z , then a sufficiently small modification retains one crossing.

Proof. Choose a small topological ball B about z in which P is a proper unknotted disk. All changes will occur away from ∂B . Straighten I in its supplied coordinates near z and choose two cap levels on opposite sides of z within a segment of the axis whose only intersection with P is z . In a parameter disk for P choose round disks $D_1 \subseteq \text{int } D_0$ about the parameter of z so that $P(D_0)$ is compactly contained in the open slab between these levels and in the straight-axis chart. Now choose a sufficiently thin polygonal cylinder C about the intervening axis interval. Its caps miss P . Compactness and the isolated-axis intersection make all intersections with its side annulus A lie in $P(D_1)$ once the radius is sufficiently small. Every side loop and its bounded parameter subdisk then lie in D_1 . Thus all disks subsequently used for linking lie inside the cap slab and cannot meet the axis extended beyond the isolated supplied segment.

Put P in PL general position near A , without changing it near the axis or near ∂B . To justify this preliminary operation, write P as the image of a flat disk under a chart homeomorphism, approximate that homeomorphism by a PL homeomorphism on a compact neighborhood of the prospective side intersections, and install the approximation on a smaller closed neighborhood by Lemma 2.3. The support misses the axis, and the unsupported part of the disk has a positive gap from the cutting side. Take the perturbation sufficiently small that $P(D_0)$ remains in the open cap slab and all prospective side intersections remain in its parameter interior. A small change of the polygonal side gives transverse intersections. Consequently $P \cap A$ is a finite family of polygonal simple closed curves.

First remove every curve which is null-homotopic in A . Choose an innermost such curve on A . Its annular disk has interior disjoint from P : it contains no further null curve by the choice, and cannot contain an essential curve of A . Replace the disk which this curve bounds in P by that disk on A , pushed slightly off A . Round the seam on the side dictated by the remaining part of P . The annular disk lies inside the open cap slab; its small push can be kept there. The operation replaces a parameter subdisk of D_0 , and hence the remaining relevant subdisks still lie in that slab. It gives an embedded locally flat disk, removes the chosen intersection curve, and creates no new side or axis intersections. Repeating finitely many times leaves only essential side curves. The disks discarded in this operation may contain z ; in that event the axis intersection is simply removed.

Every remaining essential side curve has linking number ± 1 with the extended straight axis. Its disk in P must therefore meet that axis. Since the surgeries have added no axis intersections, this

disk contains the original isolated intersection. The disks bounded by the remaining curves are thus nested. Replace the outermost one by a tame proper disk inside C with the same rim and with exactly one flat transverse crossing of the axis. Such a disk is obtained by isotoping the essential rim in the side annulus to a circular section and extending that isotopy across an outer collar of a meridian disk. After this replacement the rest of P is outside the cylinder. If no side curves remain, there is no intersection with the cylinder at all: the caps miss P , the outer boundary of the proper disk lies outside C , and any component entering C would have to cross its side.

The result is a locally flat proper disk in B agreeing with the original disk near its boundary. Both disks split B into balls. The locally flat Schoenflies theorem [10, Theorem 4], followed by extension on the two sides, gives an ambient homeomorphism of B , fixed on ∂B , carrying the original disk to the constructed disk. Extend it by the identity outside B . The ball, the cylinder, and each surgery support can be chosen successively smaller, proving the support assertion.

Finally, in the opposite-side case retain two nearby axis points on opposite sides, outside the modification support. Separation forces the modified disk to meet the intervening interval. Since the construction leaves at most one intersection, exactly one remains. $\square$

Corollary 3.6 (Affine germs at cleaned crossings). *Let h be a homeomorphism between open three-manifolds. A crossing of a regular source surface with the preimage of a target line can be cleaned by applying Lemma 3.5 to the image surface. A crossing of a regular source curve with the preimage of a target surface can be cleaned by applying the Lemma to h^{-1} . If the regular strata are straightened in C^1 coordinates, the modified homeomorphism can additionally be made affine on a smaller neighborhood of every retained crossing. The only crossings there are the retained transverse ones. Supports can be prescribed to be small in both the source and target.*

Proof. After cleaning, use the indicated coordinates so that the relevant source and image surface germs are planes. Choose a small outer source ball in this germ and a much smaller inner source ball. Prescribe an affine map on the inner ball, sending its equator to the target plane, with its image contained in the outer image ball. Choose the affine map to agree with the side correspondence and orientation of h .

Extend first across the planar annulus between the two equators. On either side, the outer half-ball is a topological ball with a flat boundary patch near the equatorial center. Removing the inner half-ball, attached along a disk in that patch, again leaves a ball. This follows either from a boundary half-space chart and relative Schoenflies or from the usual ball-with-boundary-attached-ball model. The resulting boundary homeomorphisms extend across these balls. The two extensions agree on the planar annulus, so they define the required cutoff of h . The surface remains in the target plane during this further change. In the inverse construction the inverse affine germ is again affine. Finally, continuity of h and h^{-1} permits the paired support neighborhoods to be made small in both spaces. $\square$

A crossing has now been made regular without changing the map away from a small paired neighborhood. To install an identity germ at a center with several prescribed rays, we must also control the arcs between successive surrounding spheres and their boundary identifications. The next lemma identifies the individual arcs; the following product lemma identifies the shell together with all its labelled strands. These are the two topological inputs to Proposition 3.9.

Lemma 3.7 (A two-point cut of an unknot). *Let K be a tame unknot in the three-sphere and let B be a locally flat ball whose boundary meets K in exactly two standard transverse crossings. Then the proper arc $K \cap B$ is boundary-parallel in B . In particular, the ball-arc pair is the standard unknotted interval pair.*

Proof. Choose a tube about K meeting ∂B in two meridian disks. Such tubes can be constructed on the two cut intervals, starting with the product charts at their endpoints and continuing ordinary regular neighborhoods along their interiors. Glue the endpoint disks to obtain the tube about K . The exterior of the whole tube has fundamental group $\mathbb{Z}$, generated by a meridian. Its two pieces are glued along the annulus obtained from ∂B by deleting the meridian disks. The inclusion of this annulus into the whole exterior induces an isomorphism on fundamental groups. Hence its inclusions into the two pieces are injective. The injective-amalgam form of van Kampen then embeds each piece's group in the total group. Since its image contains the meridian generator, each piece has group $\mathbb{Z}$, again generated by that meridian.

In the exterior piece lying in B , join the two end circles by an arc on the annulus from ∂B , and close it by a connector on the lateral annulus of the drilled tube. This is a simple loop on the boundary of the exterior. Its class is a power of the meridian. Twisting the connector by the opposite integer power on its annulus makes the loop null-homotopic while preserving simplicity. Dehn's Lemma supplies a properly embedded disk with that boundary.

Restore the tube and attach the radial strip joining its connector to the core arc. The strip is embedded even when the connector twists: in product coordinates its angle may vary with the interval coordinate while its radius decreases to zero at the core. Its two short edges lie in the meridian end disks. The resulting locally flat disk has boundary consisting of $K \cap B$ and an arc in ∂B . Thus the arc is boundary-parallel. Sliding along a neighborhood of this disk identifies the pair with a ball and a standard diameter. Rounding the seams does not change this conclusion. $\square$

Lemma 3.8 (A sphere crossing radial strands once). *Let $0 < a < b$, let $v_1, \dots, v_n \in S^2$ be distinct, with $n \geq 3$, and put*

$$M = \{x \in \mathbb{R}^3 : a \leq |x| \leq b\}, \quad L_i = \{tv_i : a \leq t \leq b\}.$$

Suppose that $S \subset \text{int } M$ is a locally flat embedded sphere separating the two boundary spheres of M . Suppose that S meets each L_i at exactly one point z_i , and that these intersections are standard topological transverse crossings: the surface-arc pair has a local chart onto a plane and a transverse straight line. Let U be the closed region between $\{|x| = a\}$ and S . Then there is a homeomorphism of pairs

$$\Phi : (S^2 \times [0, 1], \{v_1, \dots, v_n\} \times [0, 1]) \longrightarrow (U, \bigcup_i (L_i \cap U))$$

with $\Phi(q, 0) = aq$ and $\Phi(v_i, 1) = z_i$. Moreover, for any such product parametrization, the upper boundary map $\phi(q) = \Phi(q, 1)$ satisfies

$$q \longmapsto \frac{\phi(q)}{|\phi(q)|} \simeq \text{id}_{S^2} \quad \text{relative to } \{v_1, \dots, v_n\}.$$

This homotopy takes unmarked points to unmarked points at every time. Equivalently, the link identification supplied by the product agrees, up to a homotopy preserving the marked points and their complement, with radial projection.

Proof. Write $P = S^2 \setminus \{v_1, \dots, v_n\}$ and $S^\circ = S \setminus \{z_1, \dots, z_n\}$. Radial projection restricts to

$$f : S^\circ \longrightarrow P.$$

This map is proper: its extension to the compact sphere S has $f^{-1}(v_i) = \{z_i\}$, so the inverse image of a compact subset of P is a compact subset of S° . Orient S as the boundary of its bounded complementary region. Since S encloses the inner sphere, its unpunctured radial projection has degree 1. Local degree at any point of P therefore shows that the proper map f also has degree 1.

We record explicitly the covering argument for f_* . Let $p : \tilde{P} \rightarrow P$ be the connected covering corresponding to $f_*\pi_1(S^\circ)$, and let $\tilde{f}$ be the lift of f . The lift is proper: for compact $K \subset \tilde{P}$, its inverse image is closed in the compact set $f^{-1}(p(K))$. Give the covering its induced orientation and let d be the proper degree of $\tilde{f}$. For a fixed $y \in P$, compactness of $f^{-1}(y)$ implies that its lifted image meets only finitely many points of the discrete closed fibre $p^{-1}(y)$. Additivity of local degree expresses $\deg f$ as the sum of the local degrees over these points. At every point of the covering the local-degree sum for $\tilde{f}$ equals d ; it is zero at a point outside its image. Consequently an infinite-sheeted covering would force $d = 0$ and then $\deg f = 0$. For a finite-sheeted covering with k sheets, the same calculation gives $1 = kd$. Hence $k = 1$, and f_* is surjective. Both fundamental groups are free of rank $n - 1$, so the Hopf property of finitely generated free groups [16, III.A, Complements 18–19] makes f_* an isomorphism.

Choose pairwise disjoint closed regular-neighborhood tubes N_i around the full strands L_i , each meeting S in one meridian disk and meeting each boundary sphere of M in one end disk. Such tubes are obtained from the pair charts at the crossings and regular neighborhoods of the two remaining tame subintervals, using the same end disks where they join. In particular all tubes and their pieces on either side of S are standard interval neighborhoods. Drill out their interiors relative to M , obtaining an exterior E . Denote its bottom and top planar boundary surfaces by F_0 and F_2 , and the truncated copy of S by F_1 . Regular-neighborhood coordinates identify the fundamental groups of these exteriors with those obtained by deleting just the core strands, compatibly with the inclusions of the truncated surfaces. Since

$$M \setminus \bigcup_i L_i \cong P \times [a, b],$$

our calculation of f_* shows that $F_1 \hookrightarrow E$ induces a fundamental-group isomorphism. The same is immediate for F_0 and F_2 .

Cut E along F_1 , and let W be the piece on the inner side. The inclusion of F_1 into either piece is injective on fundamental groups, because its composite into E is injective. Van Kampen and the normal-form Theorem for a free product with amalgamation therefore embed both piece groups in $\pi_1(E)$. As the common group $\pi_1(F_1)$ already maps onto $\pi_1(E)$, both piece inclusions are isomorphisms. It follows that both

$$\pi_1(F_0) \longrightarrow \pi_1(W), \quad \pi_1(F_1) \longrightarrow \pi_1(W)$$

are isomorphisms. Apart from these horizontal surfaces, the boundary of W consists of n annuli A_i , one from each drilled tube, joining the corresponding boundary circles of F_0 and F_1 .

The manifold W is irreducible. Indeed, let Σ be a locally flat sphere in its interior. It cannot separate the boundary spheres of M , since every full strand joins those spheres and is disjoint from Σ . Schoenflies therefore gives a ball B bounded by Σ and contained in $\text{int } M$. Each N_i is connected, is disjoint from Σ , and meets ∂M , so it lies outside B . The connected surface F_1 also lies outside B , since it is disjoint from Σ and its boundary meets the tubes. Thus $B \subset W$. All subsequent disks and spheres may be taken locally flat, or PL after triangulating this compact 3-manifold.

We now give a relative product recognition argument, including the boundary identifications. Orient F_0 and F_1 so that oriented peripheral loops correspond across the annuli A_i ; these are opposite to one another when compared with their induced orientations as parts of ∂W . The fundamental-group identification between them preserves the individual oriented peripheral conjugacy classes. The surface classification Theorem in its peripheral-preserving Dehn–Nielsen form supplies a homeomorphism $j : F_0 \rightarrow F_1$ realizing this outer isomorphism and matching the labelled boundary circles [18, Theorem 8.8 and Proposition 3.19].

Here is the adjustment from outer to based identifications. For this step relabel the boundary circles and their annuli by $0, \dots, n-1$. Choose a basepoint x_0 on one boundary circle C_0 of F_0 , and a connector Δ_0 in its annulus from x_0 to $j(x_0)$. Transport the second inclusion into W along Δ_0 . Its composition with j_* and the first inclusion differ by an inner automorphism. Both carry the positive based peripheral of C_0 to the same element g_0 of $\pi_1(W, x_0)$, so the conjugating element centralizes g_0 . Every peripheral of an n -holed sphere is primitive in its free fundamental group; since $n \geq 3$, its centralizer is precisely the cyclic group it generates. Twisting Δ_0 by an appropriate integer power in its annulus therefore gives the based equality

$$[c] = [\Delta_0 j(c) \Delta_0^{-1}] \quad \text{in } \pi_1(W, x_0) \quad \text{for every based loop } c \text{ in } F_0. \quad (*)$$

Choose disjoint properly embedded arcs $\alpha_1, \dots, \alpha_{n-1}$ in F_0 , joining C_0 to the other boundary circles, whose cutting opens F_0 to a disk. Their endpoints on C_0 are distinct. For each other circle choose the endpoint x_i of its arc, a path c_i from x_0 to x_i following C_0 and then α_i , and an initial connector Δ_i in A_i from x_i to $j(x_i)$. The two paths

$$c_i \Delta_i \quad \text{and} \quad \Delta_0 j(c_i)$$

transport the peripheral at $j(x_i)$ to the same based element of $\pi_1(W, x_0)$, by $(*)$ and the annulus correspondence. Their discrepancy thus centralizes that peripheral. The same centralizer argument allows an integer twist of Δ_i making these two paths homotopic relative to their endpoints in W .

Parametrize each A_i as a circle product, with its lower boundary the identity, its upper boundary the restriction of j , and its chosen connector in the adjusted relative homotopy class. Such a parametrization exists because its remaining integer winding is changed by a Dehn twist of the annulus. On the base annulus use the parametrization with track Δ_0 at x_0 and use its tracks also at every endpoint of an α_i on C_0 . Thus all connectors are disjoint. The product tracks are compatible with travel along a boundary circle, so the preceding path equalities imply that each closed curve formed from α_i , its two connector tracks, and $j(\alpha_i)$ is nullhomotopic in W . These curves, denoted γ_i , are disjoint simple curves on ∂W .

Dehn's Lemma supplies properly embedded disks $D_i \subset W$ with $\partial D_i = \gamma_i$ [35]. They can be chosen disjoint: put them in general position and remove their intersection circles by the usual innermost-disk replacements, keeping their already disjoint boundaries fixed. The chosen horizontal and annular maps give a boundary homeomorphism from $\partial(F_0 \times [0, 1])$ to ∂W carrying the boundaries of the rectangles $\alpha_i \times [0, 1]$ to the γ_i . Consequently cutting W along the D_i produces a manifold whose boundary is one sphere, just as cutting $F_0 \times [0, 1]$ along those rectangles does. Every component of a manifold obtained by these proper disk cuts has nonempty boundary, so the cut manifold is connected. Irreducibility persists under the cuts: a sphere in the cut manifold bounds a ball in W , and each cutting disk, being connected and having boundary on ∂W , must lie outside that ball. A collar inset of the spherical boundary then shows that the cut manifold is a ball.

Extend the boundary correspondence over the rectangles and the D_i , and then over the resulting balls. Regluing gives a product homeomorphism $F_0 \times [0, 1] \rightarrow W$ realizing the chosen boundary correspondences. Restore each removed tube piece. Extend its boundary-circle maps over its two cap disks, taking the marked core endpoints to one another. A homeomorphism of the boundary of a standard ball-interval pair preserving the two endpoints extends to the pair: identify both pairs with a ball and a diameter and cone the boundary map. Applying this to each tube extends the exterior product to the asserted product of the filled region and its labelled strands. At the lower end choose the cap extensions to agree with the identity, so that $\Phi(q, 0) = aq$.

Finally the formula

$$(q, t) \mapsto \frac{\Phi(q, t)}{|\Phi(q, t)|}$$

is the claimed boundary-comparison homotopy. It fixes each v_i because its product track is contained in L_i , and it avoids all marked directions whenever q is unmarked because the product homeomorphism is a homeomorphism of pairs. $\square$

Proposition 3.9 (Replacement at a star center). *Use centered Euclidean coordinates in the source and target of a local homeomorphism h , with $h(0) = 0$. Let*

$$D = \{v_1, \dots, v_n\} \subset S^2, \quad n \geq 3, \quad R_i = \{tv_i : t \geq 0\},$$

be distinct labelled directions, partitioned into forward and reverse sets F and E . Each nonempty set has at least two elements. Choose narrow round cones C_i about R_i , and slightly wider round cones C_i^+ with pairwise disjoint angular closures. Assume, as germs at the origin, that

$$h(R_i) \subset \text{int } C_i \quad (i \in F), \quad h^{-1}(R_i) \subset \text{int } C_i \quad (i \in E).$$

For each nonempty distorted system, assume there are nested round spheres with radii tending to zero in its own space, each meeting every arc of that system exactly once, with the arc passing from one side of the sphere to the other. Impose one of the following angular tests.

1. *There is a finite based generating loop system $(\gamma_1, \dots, \gamma_{n-1})$ for $S^2 \setminus D$, chosen outside the closed angular caps of all the cones. At arbitrarily small common source radii r , all parametrized loops*

$$s \mapsto \frac{h(r\gamma_a(s))}{|h(r\gamma_a(s))|}$$

are close to γ_a within a fixed simultaneous homotopy tolerance in $S^2 \setminus D$. The tolerance also keeps the image loops outside the cone caps. Their common basepoint is identified with the original basepoint through the small basepoint neighborhood in this tolerance.

2. *There are only forward directions, all lying cyclically on a source equator and corresponding to the same target equator directions. The map h preserves orientation. At arbitrarily small source radii the parametrized radial projection of the image of that equatorial circle is uniformly close to its original parametrization, with tolerance small compared with the separations of successive marked points.*

The cones are taken narrow enough relative to these tolerances and separations. Then h can be changed in arbitrarily small paired neighborhoods of the center to equal id as a germ. The forward and reverse angular conditions persist, with the margins from C_i to C_i^+ if necessary. Any additional strict safety conditions away from the germ are preserved by making the supports sufficiently small. The assertion imposes no identity condition at other points of the original intervals.

Proof. We first straighten the distorted arc systems within their separate angular sectors. Each distorted arc is tame as an ambient arc, being the image or preimage of a straight interval under a homeomorphism. At each of its intersections with a chosen round section, straighten the arc in an ambient chart and apply Lemma 3.5 to the section disk. The opposite-side crossing persists once if the support is sufficiently small. Move the changed section back to the round section by the inverse supported homeomorphism, dragging the arc with it. The resulting arc has a standard transverse crossing with the original round section. Choose these changes in disjoint annular patches inside the relevant angular sectors. They can be made arbitrarily small and preserve nesting of the sections. All section crossings are now standard pair crossings.

Consider the connecting strand between two successive sections inside one angular sector. Their annular-sector region is a ball, and this strand is a proper interval in it. It is unknotted for the

following reason. Choose a second arc of the same distorted system, which is possible by the cardinality hypothesis. On the undistorted side of the system, the two radial arcs can be closed outside a smaller germ neighborhood by a polygonal path to give an unknot. Take the closing polygon and its spanning disk in a sufficiently small original chart ball whose image remains in the working chart. The image of the closed curve is still an ambient unknot: the chart-ball homeomorphism extends across its exterior by locally flat Schoenflies. For a reverse system use h^{-1} in this argument. Choose the round sections so close to the center that they avoid the closing path. The second arc lies in its own disjoint sector, so the first sector's annular ball meets this unknot in just the strand under consideration, with exactly two boundary crossings. The section-cleaning homeomorphisms preserve the ambient unknot. Lemma 3.7 therefore makes the strand boundary-parallel in its annular-sector ball.

Straighten each such strand in that ball, using matching maps on the cap disks and the identity on the sector sides. The prescribed cap maps present no extra obstruction: for the standard ball-diameter pair, a homeomorphism of the boundary sphere respecting the two ends extends by coning, and fixes the diameter as a set. At the outermost section make a transition inside the same sector and extend by the identity outside a slightly larger neighborhood. The infinitely many annular straightenings glue to a homeomorphism at the center, with continuous inverse there, because their supports lie in balls with radii tending to zero.

Perform the target changes for the forward system and the source changes for the reverse system. All the sectors are disjoint, so the target changes leave the target reverse rays fixed, and the source changes leave the source forward rays fixed. Pre- and post-composing accordingly gives a homeomorphism H carrying every labelled ray of the combined system to itself as a set, as a germ. The generator test is unchanged: its source loops and their image directions have positive margins from the supports. In the equatorial case the target changes have arbitrarily small angular displacement, since they act in sufficiently narrow sectors; thus the parametric angular test persists within its homotopy tolerance.

Choose a test radius r inside this exact-ray germ. The locally flat sphere $S = H(\partial B_r)$ meets each marked ray once in a standard topological crossing. Indeed H carries the standard source sphere-ray pair to this pair. A sufficiently small round target sphere lies inside $H(B_r)$; together with S it bounds a shell to which Lemma 3.8 applies. Its product comparison of the marked boundary spheres is homotopic to radial projection on their punctured complements.

In the first angular case the comparison of the outside boundary map with the intended identity map induces the identity on the chosen based generators. The punctured-sphere mapping-class Theorem implies that it is isotopic to the identity through maps fixing the labelled marked points [18, Theorem 8.8]. The same Theorem detects the orientation here; the assumption $n \geq 3$ excludes the two-puncture ambiguity.

In the equatorial case take a consecutive spanning chain of equatorial joining arcs between marked points, omitting one closing arc. Their projected images represent the same classes relative to their marked endpoints. To see this directly, the middle of each parametrized image lies in a narrow corridor containing no puncture, while each short end lies in a disk containing only its own marked point. In such an end disk, possible angular winding can be interpolated using lifted polar angles on the open end of the arc; the radius tends to zero at the endpoint throughout the interpolation. Thus no framing obstruction remains at that marked point. Interior points of the image arcs avoid all marked points because H already maps the complete labelled ray germs exactly to themselves. The homotopy comparison in Lemma 3.8 transfers these arc classes to the boundary identification. The usual disjoint-arc isotopy test fixes these classes simultaneously [18, Section 1.2.7, Proposition 2.8, and Lemma 2.9]. Cutting along this chain leaves a disk, on which the orientation-preserving boundary identification is isotopic to the identity. This proves the required

trivial marked mapping class also in the second case.

Choose a smaller source sphere ∂B_s whose intended identity image is contained in $H(B_r)$. Product coordinates on the source annulus and on the target annulus, together with the preceding marked-sphere isotopy, extend the outside values of H and the inside identity across the annulus while respecting every labelled ray. Set the map equal to the identity on B_s and equal to H outside B_r . This is a homeomorphism and supplies the asserted identity germ. The earlier sector straightenings and the final cutoff have arbitrarily small support in the corresponding spaces. Their angular bounds and the given strict margins prove the remaining assertions. $\square$

Lemma 3.10 (Relative regularization with protected germs). *Suppose a homeomorphism has been made affine in specified C^1 coordinates near a disjoint locally finite family of crossing points, and has prescribed exact affine or identity germs at finitely many star centers. It admits a fine approximation by a locally bi-Lipschitz homeomorphism retaining smaller exact germs. On each compact the approximation is given by finitely many C^1 formulas on closed pieces. Positive-gap avoidance conditions, the retained transverse crossing counts, and strict angular conditions can be preserved. Fine approximation tolerances can also impose reciprocal closeness on compact sets.*

Proof. First realize the required coordinate changes by supported C^1 diffeomorphisms. After separating an affine first jet, such a local coordinate change has the form $g(0) = 0$, $Dg(0) = I$. On a sufficiently small ball, a cutoff

$$\tilde{g}(x) = x + \chi(x)(g(x) - x)$$

equals g on a smaller ball and the identity outside the larger one. Because $Dg - I = o(1)$ and $g(x) - x = o(|x|)$ at the center, its derivative is uniformly close to I when the balls are sufficiently small. It is therefore a supported C^1 diffeomorphism. Use disjoint locally finite support neighborhoods in the source and target; the affine parts are absorbed into the prescribed affine germs. After these coordinate changes the homeomorphism is literally affine on neighborhoods of all protected points.

Enclose smaller germs by closed polyhedra inside those affine neighborhoods and apply the relative PL approximation of Lemma 2.2. Equivalently, excise these neighborhoods and approximate on their open complement as a PL three-manifold with boundary, relative to its already prescribed boundary collar. Only finitely many protected pieces occur on any compact, so the construction is locally finite. Undoing the supported C^1 coordinate changes gives a locally bi-Lipschitz homeomorphism with the stated finite piecewise- C^1 description on compacts.

The exact germs preserve the crossings there. Elsewhere the specified avoidances have positive gaps on the compact pieces where they are needed, so sufficiently fine approximation creates no additional intersections. The same choice preserves angular inequalities with a strict margin. Finally, for a homeomorphism fine source-dependent approximation tolerances can be chosen to ensure prescribed inverse tolerances on target compacts; use continuity of the original inverse and compact exhaustion. $\square$

We can now regularize finitely many crossings on each compact set and retain exact germs at the protected centers. The resulting map is locally bi-Lipschitz, but its derivative bounds may be arbitrarily large. The next construction therefore estimates selected image curves and surfaces: generic sampling controls their intersection counts, and a target reparametrization converts those counts into length and area.

3.4 Generic traces and target triangulations

Lemma 3.11 (Generic trace properties). *Let $f : \Omega \rightarrow \Omega'$ be a homeomorphism of open subsets of $\mathbb{R}^3$ with $f \in W_{\text{loc}}^{1,p}$, $1 \leq p \leq 2$. Consider locally finite families of compact Lipschitz semialgebraic*

parametrizations of dimension $m = 1$, and also of dimension $m = 2$ if $p = 2$. Almost every translation of each template can be chosen so that its f -image is countably m -rectifiable and has the area formula with tangential derivative, counted with the parametrization's multiplicity. The same conclusions hold on all lower strata needed in a finite stratification. At a generic intersection with a complementary-dimensional affine test stratum, the source trace has a regular manifold stratum and regular full-rank parameter branches, with locally constant multiplicities. Exceptional image values, including branch junctions and dimension-drop intersections, may be excluded from such intersections.

Proof. For $m = 1$, Sobolev slicing gives absolute continuity on almost every parallel line, and hence the one-dimensional area formula and Lusin property. A smooth curve chart can be foliated by transverse translates and straightened by a local ambient smooth diffeomorphism, giving the same conclusion in that chart. For $m = 2$ use [14, Theorem 1.3]: a $W^{1,2}$ homeomorphism in dimension three has the two-dimensional Lusin property on almost every parallel plane. Together with Sobolev slicing and the usual Lipschitz decomposition of a Sobolev map [8, Theorem 3(1)], this gives the surface area formula. Apply the Csörnyei–Hencle–Malý theorem after straightening a smooth graph by an ambient smooth diffeomorphism. Fixing any other translation coordinates and then applying Fubini proves the assertion for the translated graph templates.

Stratify the semialgebraic source parametrizations $P : K \rightarrow \Omega$, their source images $P(K)$, and the ranks and overlaps of these parametrizations, using a common finite refinement as in [20, Proposition 2.3 and Lemma 2.5]. This stratification concerns P , not the generally non-semialgebraic map $f \circ P$. On a full-rank branch the preceding graph argument applies, and its parameter derivative is $Df(P)DP$. On a lower-rank piece, use the slicing assertion for its lower-dimensional image stratum; for instance a one-dimensional image is rectifiable by the curve assertion and has zero two-dimensional measure. These observations also make images of parameter-null sets negligible on the regular branches. The chain rule and tangential integrability may alternatively be obtained by approximating f by smooth maps in ambient Sobolev norm and using Fubini to pass along a subsequence in parameter Sobolev norm.

We use the multiplicity area formula and the rectifiable-set coarea formula in [37, Chapter 2, Section 3, and Chapter 3, Section 2]. The area formula and translation coarea imply that almost every complementary affine test avoids the exceptional image sets. At a remaining source value, stratification, local inversion on regular branches, and compactness leave a single local smooth source stratum with a fixed finite number of parameter branches. Dimension-drop intersections of different pieces are negligible by the same argument. More explicitly, full-rank strata of the finite refinement are diffeomorphic onto their images, which can be made identical or disjoint; their number, rather than compactness alone, bounds the branch multiplicity. On relatively compact branch charts P is bi-Lipschitz, so parameter-null sets first have null source image and then null target image by the separate sliced Lusin property. Finally, $p \geq m$ in the admitted cases. Translation and Fubini give finite tangential m -mass on almost every compact template, and the area formula for its orthogonal projection makes the complementary intersection count finite for almost every affine test. The locally finite family involves only finitely many templates on each compact set; intersect their full-measure choices and then use a countable exhaustion. This proves all the simultaneous assertions. $\square$

We describe the target meshes used below. Their shape need not be uniform; only bounded local site counts and upper size bounds are needed for target budgets. Uniform source shapes will be obtained separately in Lemma 3.20.

Lemma 3.12 (Target mesh with lattice patches). *In an open subset $V \subset \mathbb{R}^3$, prescribe finitely many separated compact patches with buffers. In a deep part of each patch one may require a*

rotated rectangular lattice of side lengths d or ηd , where $0 < \eta \leq 1$ is fixed. There are locally finite triangulations of V agreeing there with the chain, or Kuhn, triangulation of that lattice and with arbitrarily small prescribed local upper size bounds. For each local candidate, its sampling variables admit a physical common translation t and normalized relative coordinates z whose joint density satisfies

$$q(t, z) \leq Cd^{-3}, \quad t \in B(t_0, Cd), \quad z \in Z \subset B(0, C), \quad (3.10)$$

where the dimension and measure of Z are bounded. Deterministic lattice relative coordinates are held fixed; when there are no free relative coordinates their integration means a unit point mass. The bounds are universal away from enlarged lattice patches and may depend on η inside them. Only boundedly many candidates occur within a bounded number of local mesh lengths. Formula (3.10) is used by joint integration, without normalizing a conditional translation law after fixing z .

Proof. Choose a positive size function $d(u)$ with sufficiently small absolute Lipschitz constant, satisfying the prescribed local upper bounds and $d(u) \ll \text{dist}(u, \partial V)$. Make it constant, with the desired common value, near each lattice patch. Outside deep lattice regions take a maximal packing at this scale, then independently jitter each retained site in a ball of a fixed small fraction of its local scale, with uniformly bounded density. In a lattice patch use a common uniform positional shift over one period. Truncate the unshifted lattice lists in their buffers, leaving overlap with the background cloud; the separated lattice patches remain many steps apart. The resulting sites cover at distance at most $Cd(u)$ and have bounded counts within that distance. A minimum separation between every pair of sites is unnecessary.

Use their Delaunay subdivision, with a fixed pulling order to triangulate nonsimplicial cells. Its localization can be checked inside the open domain. An empty ball containing or passing through a point near u cannot have radius much larger than $d(u)$: moving several mesh lengths from that point toward the center stays inside V and gives a location well inside the ball; the covering property then puts a site strictly inside it. The slow variation of d justifies using the same scale during this argument. Finite restrictions of the cloud therefore stabilize near u once they contain a sufficiently enlarged local cloud. Their ordinary Delaunay subdivisions cover and agree there. This proves local existence, local finiteness, and compatibility of the subdivisions on V . Every candidate uses a bounded list of neighboring sites at comparable scales; coincidences occur with probability zero.

Deep in a rectangular lattice, the Voronoi description shows that the Delaunay cells are the rectangular boxes. Increasing lexicographic pulling gives their Kuhn simplices. Taking the scale small enough places any prescribed deep compact set and all its touching simplices within this region before sampling.

For the density assertion, first fix a candidate tuple of site indices, before imposing its Delaunay selection event. If it contains lattice sites, they all belong to one shifted lattice. Use its physical shift as t and write each unstructured site as $x_0 + t + dz_j$, with x_0 a fixed candidate anchor. The original joint law of the shift and the k unstructured sites has density at most $Cd^{-3(k+1)}$. Replacing the latter positions by z_j has Jacobian d^{3k} , giving (3.10). If no lattice site is present, use one site as $x_0 + t$ and express the other $k - 1$ sites relative to it in units d ; the Jacobian $d^{3(k-1)}$ gives the same bound from the original density Cd^{-3k} . The normalized relative coordinates lie in a bounded set because all candidate sites have comparable scales and lie in one bounded-scale neighborhood. The Delaunay selection event is an indicator bounded by one and is discarded for an upper estimate. All subsequent calculations integrate the unnormalized translation slice for each z , then integrate z over this bounded set. Thus no bound on a normalized conditional translation density is needed. $\square$

Counting intersections with complementary cells is also central to polyhedral deformation; see [40, Proposition 2.2]. Here the weights count unsigned parameter visits, and the concentration will

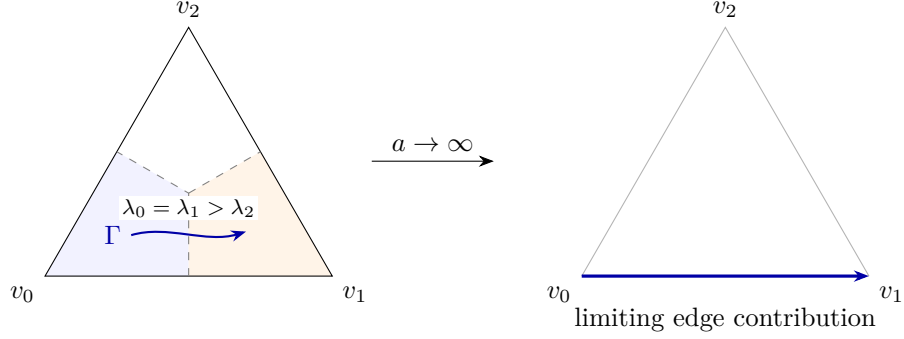


Figure 1: A two-dimensional model of a tie budget. Dashed segments are the two-leader ties. The curve crosses one of them away from the three-leader center; its image concentrates on the corresponding edge. The right panel records the limiting contribution, not the image under a finite-parameter homeomorphism.

be realized by homeomorphisms of the target.

Definition 3.13 (Tie budget). In a tetrahedron σ with vertices $v_0, \dots, v_3$, write its barycentric coordinates as λ_i . For a set I of $m + 1$ vertices, the associated dual stratum is

$$Z_{\sigma, I} = \{\lambda_i = \max_j \lambda_j \ (i \in I), \quad \lambda_j < \max_i \lambda_i \ (j \notin I)\} \cap \text{relint } \sigma.$$

Write a source parametrization as $P : K \rightarrow \Omega$ and its measured image as $\Gamma = f \circ P$; lengths and areas of Γ mean integrals on K . For such a generic measured m -trace, its budget $\mathcal{B}_m(\Gamma)$ is the sum, over its visits to these strata, of

$$\mathcal{H}^m(\text{conv}\{v_i : i \in I\}),$$

with visits counted with parameter multiplicity. A weighted budget uses in addition the supremum of a prescribed nonnegative continuous weight w over σ . A fixed linear change in the output may also be applied when computing the flat weights.

A triangle gives a simple model for this budget rule. Let v_0, v_1, v_2 be its vertices and let $\lambda_i > 0$ be interior barycentric coordinates. For a constant $a > 0$, consider the map

$$\sum_{i=0}^2 \lambda_i v_i \mapsto \frac{\sum_{i=0}^2 \lambda_i e^{a\lambda_i} v_i}{\sum_{i=0}^2 \lambda_i e^{a\lambda_i}}.$$

As $a \rightarrow \infty$, a point with one largest coordinate moves toward that coordinate's vertex. A regular curve in a sufficiently small neighborhood of a point of $\lambda_0 = \lambda_1 > \lambda_2$, crossing that tie once transversely with endpoints in the two adjacent one-leader regions, instead runs between v_0 and v_1 . The concentration calculation below shows that one such crossing contributes the edge length $|v_1 - v_0|$ in the limit; Figure 1 illustrates this local situation. In a tetrahedron the same principle pairs a curve with a two-leader surface and an edge weight, or a surface with a three-leader line and a triangle weight. This is why the budget uses complementary dimensions and flat primal faces.

For generic samples all relevant hits are finite on compact measured pieces. Higher ties, simplex-boundary junctions, and lower-dimensional source exceptions are avoided by Lemma 3.11 and translation coarea. Counts consequently concern isolated hits at regular underlying source strata; no differentiability of the original image surface at such a hit is being assumed.

3.5 A variable simplex squeeze

Lemma 3.14 (Compatible simplex squeezing). *Let $\mathcal{T}$ be a locally finite triangulation of a three-dimensional open domain. Assign constants $0 < b_\sigma \leq 1$ to its tetrahedra, and to every nonempty face ρ assign the minimum of the constants of its incident tetrahedra. On each tetrahedron put*

$$b(\lambda) = \min_{\emptyset \neq \rho \subseteq \sigma} \left(b_\rho + \frac{\max_{i \notin \rho} \lambda_i}{\max_j \lambda_j} \right), \quad (3.11)$$

$$a(\lambda) = e^{1/b(\lambda)} - e, \quad T_i(\lambda) = \frac{\lambda_i e^{a(\lambda) \lambda_i}}{\sum_j \lambda_j e^{a(\lambda) \lambda_j}}, \quad (3.12)$$

where the maximum over an empty set in (3.11) is zero. These formulas define a face-preserving, locally bi-Lipschitz self-homeomorphism T of the domain, with finitely many C^1 formulas on closed pieces of every compact subset. The normalized Lipschitz constant of b is bounded absolutely, independently of its minimum, and a.e.

$$|Da| \leq C(a+1)(1+\log(a+1))^2. \quad (3.13)$$

If all the incident constants in a mesh neighborhood are one, T is the identity on its inner simplices.

Let G parametrize finitely many compact m -traces, $m \in \{1, 2\}$, by finite closed-piece C^1 formulas, and suppose their intersections with the $Z_{\sigma, I}$ are regular transverse hits away from higher ties and simplex boundaries. Every counted hit is in the interior of each counted regular m -dimensional parameter branch; parameter boundaries and lower-dimensional parameter strata avoid the counted dual strata. As the lower bounds of a tend to infinity on the measured neighborhood,

$$\int J_m D(T \circ G) \longrightarrow \sum_{\text{hits on } Z_{\sigma, I}} \mathcal{H}^m(\text{conv}\{v_i : i \in I\}), \quad (3.14)$$

with multiplicity. The local convergence, and in particular its upper bound with arbitrarily small error, is uniform when other floors of a are made arbitrarily larger. Continuous weights may be bounded by their simplex suprema in this conclusion.

Proof. Since the candidate $\rho = \sigma$ gives $b \leq b_\sigma \leq 1$, $a \geq 0$. Every denominator $\max \lambda_j$ is at least $1/4$, so the normalized Lipschitz constant of b is absolute. The chain rule gives (3.13). On a face τ , replacing a candidate ρ by $\rho \cap \tau$ leaves its nonzero outside coordinates unchanged and can only lower its constant, because $b_{\rho \cap \tau} \leq b_\rho$. An empty intersection gives an outside-coordinate ratio of one and cannot improve on $b_\tau \leq 1$. Thus restrictions agree on faces. Positivity, zero coordinates, and the order of coordinates are preserved by (3.12).

We verify invertibility even though a is variable. Given output ratios $z_i = T_i/T_{\max} \in [0, 1]$, for a trial $D \geq 0$ let r_i be the unique solution of

$$z_i = r_i e^{-D(1-r_i)}, \quad 0 \leq r_i \leq 1. \quad (3.15)$$

The function of r_i on the right is strictly increasing. Each r_i is nondecreasing in D , with the zero and maximal ratios fixed at zero and one. Put $R = \sum_i r_i$ and $\lambda_i = r_i/R$. The trial value of a is DR , an increasing function of D with increase at least that of D . In (3.11) the outside-coordinate ratios are exactly r_i ; hence b at this trial inverse is nondecreasing in D . Its prescribed value $e^{1/b} - e$ is nonincreasing. The equation

$$DR = e^{1/b(\lambda)} - e$$

has a unique solution, by continuity, its sign at $D = 0$, and its positive sign for large D . This proves bijectivity. On an individual closed simplex b has a positive lower bound, so the solution has bounded D . Equation (3.15) has uniformly Lipschitz inverse in this bounded range, including at zero output coordinates. The scalar equation's left-minus-right increase is at least the increase in D , so its root depends Lipschitz-continuously on the output. Since $T_{\max} \geq 1/4$, forming the output ratios is Lipschitz as well. The forward formula is Lipschitz on that simplex, and its inverse is Lipschitz. The finite min/max alternatives in (3.11) give finitely many closed pieces with C^1 extensions of the formulas. Local finiteness proves all the global assertions. If the incident constants are one, the minimum is one, so $a = 0$ and $T = \text{id}$.

It remains to prove the concentration assertion. On a compact trace set away from $(m + 1)$ -fold leader ties, at most m coordinates can be near the maximum. Coordinates outside that set and all their derivatives are exponentially small in a , multiplied only by powers of a and $\log(a + e)$ by (3.13). The active coordinates have total mass one up to those errors and therefore take values in a face of dimension at most $m - 1$. Their m -Jacobian vanishes. Expanding the actual m -Jacobian then shows uniform convergence to zero away from the counted ties, including along simplex faces by the same piecewise calculation.

Near a transverse hit relabel the leaders $0, \dots, m$ and use

$$u_i = \lambda_i - \lambda_0, \quad 1 \leq i \leq m,$$

as trace coordinates. All leader coordinates have a fixed positive lower bound there, and the remaining coordinates are separated below them. Normalize the probabilities on the leaders first. Their logit differences are

$$a(u)u_i + \log \frac{\lambda_i(u)}{\lambda_0(u)}.$$

Inactive probabilities and derivatives are exponentially small. Write $a_0 = a(0)$. Since $b(0) = 1/\log(a_0 + e)$ and b is uniformly Lipschitz, on $u = t/a_0$ with bounded t one has

$$\log \frac{a(u) + e}{a_0 + e} = O\left(\frac{(\log(a_0 + e))^2 |t|}{a_0}\right) = o(1).$$

The scaled derivative terms coming from Da tend to zero by (3.13). The leader map and its a.e. derivatives therefore converge locally uniformly, apart from its harmless piece seams, to

$$t \mapsto \left(\frac{e^{t_i}}{1 + \sum_{j=1}^m e^{t_j}} \right)_{i=1}^m.$$

This softmax map is a diffeomorphism from $\mathbb{R}^m$ onto the interior of the standard m -simplex. Multiplication by the physical edge vectors consequently gives exactly the flat m -area of the leader face, provided the tails vanish.

Here are uniform tail bounds. The determinant of the active probability derivative is its probability product times the determinant of the logit derivative. The logit derivative has the form $aI + u \otimes Da + B$, with B bounded for the fixed trace chart. The probability product is at most $Ce^{-ca|u|}$. Thus the absolute determinant, modulo exponentially small inactive terms, is bounded by

$$C(1 + a + |Da||u|)^m e^{-ca|u|}. \quad (3.16)$$

For $|u| \leq a_0^{-1/2}$, the preceding logarithmic estimate is uniform and gives $a/a_0 \rightarrow 1$ there. After scaling, the determinant has an integrable bound $Ce^{-c|t|}$, because $|Da||u|/a_0 \leq C(\log a_0)^2 a_0^{-1/2} = o(1)$. For

$$a_0^{-1/2} < |u| \leq c_1/\log(a_0 + e),$$

choose c_1 small relative to the fixed Lipschitz bound for b . Then $a \geq a_0^{3/4}$ for large a_0 , and $a|u| \geq a_0^{1/4}$. Exponential decay in (3.16) absorbs all polynomial and logarithmic factors uniformly for every larger a . In the remaining sufficiently small neighborhood,

$$b(u) \leq b(0) + C|u| \leq C'|u|, \quad a(u) + e \geq e^{c'/|u|}.$$

For small $r = |u|$, the expression $a^N e^{-car}$ is decreasing once $a \geq e^{c'/r}/2$, for any fixed N . Increasing N to absorb the logarithms in (3.16) therefore gives an integrable bound of the form

$$C \exp(Nc'/r) \exp(-c''re^{c'/r}),$$

independent of how high the local floors are. At every fixed $u \neq 0$ the integrand tends to zero as the floor tends to infinity. Dominated convergence treats this last region. These three bounds prove tail vanishing and (3.14).

For a compact measured list there are only finitely many charts and hits. One may therefore prescribe the floors simultaneously, with any allocated small errors. Local finiteness then permits such choices for a locally finite list, including summable errors. The map moves a point only inside its own simplex, so multiplication by simplex suprema of continuous weights proves the weighted upper bound. The same reasoning applies to every regular parameter branch, and hence respects multiplicity. $\square$

3.6 Transfer and selection of the budgets

Proposition 3.15 (Topological transfer of trace budgets). *Fix generic trace data as in Lemma 3.11 and a sufficiently fine generic mesh from Lemma 3.12. There is an onto locally bi-Lipschitz homeomorphism $H_0 : \Omega \rightarrow \Omega'$, locally specified by finitely many C^1 formulas on closed pieces, whose image lengths and areas on the measured traces are at most their tie budgets plus arbitrarily small prescribed local errors. Nonnegative continuous output weights are allowed, with the simplex-supremum budgets of Definition 3.13.*

The map can be arbitrarily close to f , with reciprocal closeness on interior compact sets, within the mesh displacement and chosen local modification errors. It can retain smaller exact affine germs previously installed at a locally finite family of crossings and finitely many star centers, and can retain strict compact forward and inverse safety conditions, including angular cone conditions, whenever they have positive margins. For this relative usage, measured data are separated from buffered open exemptions around the protected germs: a measured trace either avoids the exempt cores, or its weight vanishes there with a buffer. Every germ required to remain exact in H_0 is contained in one of these buffered exemptions. The affine crossing germs installed at the newly cleaned measured hits are retained only in the intermediate map G for the concentration calculation; no affine-germ condition at those hits is imposed on $T \circ G$. The mesh and all measured configurations are fixed before the squeezing floors are selected.

Proof. Each area-line hit is an isolated intersection of the image of a regular source surface with an affine target interval. Apply Lemma 3.5, then Corollary 3.6, in a paired source and target neighborhood. For a curve-plane hit, interchange source and target and clean the preimage plane. The supports can be chosen disjoint and locally finite. Area-line hits avoid curve-plane hits and the measured curve images; curve-plane changes can avoid all area-test lines. If parametrizations share a regular source stratum at a hit, the operation is performed once on that stratum and its fixed multiplicity is retained. Lower strata and other sheets are absent there by genericity. A surviving hit remains in its intended source stratum and target tie simplex, and the cleaning does not increase its count.

Use Lemma 3.10 to obtain a locally bi-Lipschitz map G with finite closed-piece C^1 formulas, keeping smaller exact crossing and center neighborhoods. Take the approximation smaller than the positive gaps from all forbidden strata off those neighborhoods. The measured intersections of G are now regular transverse intersections and their counts do not exceed the original budget counts. Choosing all supports and approximation errors finely also retains the stated compact forward and inverse conditions. These choices use only topology and qualitative approximation; no derivative bound for G is charged to a budget.

Apply Lemma 3.14 in the target and set $H_0 = T \circ G$. It has the asserted local regularity and is an onto homeomorphism. On each compact measured list choose the floors high enough that the image measurements cost at most the flat weights plus their allocated errors. Finitely many such lists meet each simplex, so the uniformity in arbitrarily higher floors permits simultaneous choices for the locally finite family.

For the relative assertion, first keep each protected-germ modification deep inside its exempt region. Make the mesh so fine that all simplices touching the protected no-movement core, and their needed neighbors, lie inside that exemption. Assign the value one to their b_σ constants; then $T = \text{id}$ on a smaller neighborhood of the germ. All simplices required for measured points, including points where the final weight is nonzero, remain buffered away from these cores and use the original f -crossings. Subsequent cleaning supports are disjoint from the core neighborhoods. Every motion by T stays inside a simplex. Thus the chosen fine mesh retains the strict margins and the reciprocal compact accuracies, and $w(TG(x)) \leq \sup_\sigma w$ whenever $G(x) \in \sigma$. This proves the weighted estimate as well. No unweighted estimate through an exempt affine-star core is asserted or needed. $\square$

Lemma 3.16 (Expected budgets). *For an m -trace with finite image measure, the meshes in Lemma 3.12 satisfy*

$$\mathbb{E} \mathcal{B}_m(\Gamma) \leq C \int_K J_m D(f \circ P), \quad (3.17)$$

with parameter multiplicity understood. The constant is universal outside enlarged lattice patches and may depend on their fixed aspect ratios inside them. The estimate holds locally with comparable enlargements. For continuous weights it holds with the weighted trace integral and an arbitrarily small enlargement error as the local mesh sizes tend to zero. Fixed linear output normalizations may be included in all areas.

Proof. Use the joint variables (t, z) of (3.10) for a candidate tetrahedron. For each fixed z , its dual $(3 - m)$ -piece Z_z has diameter $O(d)$ and measure at most Cd^{3-m} ; translation by t produces the actual dual piece. If $N(t, z)$ is its intersection count with the measured trace in the candidate's fixed enlarged neighborhood, translation coarea gives

$$\int N(t, z) dt \leq Cd^{3-m} \text{ local trace image } m\text{-measure}.$$

Indeed this is the area formula for the difference of the trace and dual-piece parametrizations, whose angle factor is at most one. The joint density bound therefore gives, for every such z ,

$$\int q(t, z) N(t, z) dt \leq Cd^{-m} \text{ local trace image } m\text{-measure}.$$

These are unnormalized translation integrals. Integrating z over its bounded normalized set only changes the constant. The flat m -face weight is at most Cd^m for every z , so it cancels the remaining scale. Discard the Delaunay selection indicator for this upper estimate. Only boundedly many candidate tuples occur locally and their enlargements have bounded overlap, proving (3.17). The same coarea calculation gives generic avoidance of negligible sets and finiteness of compact counts.

A continuous weight's supremum on each vanishing simplex differs from its value at a measured point by its local modulus of continuity. On a compact set multiply that modulus by the finite trace measure; a locally finite exhaustion and allocated errors give the weighted assertion. $\square$

Lemma 3.17 (Length budgets with arbitrarily high probability). *For any compact rectifiable measured curve Γ of finite length, any $\delta > 0$, and any $e > 0$, sufficiently small local mesh upper bounds make*

$$\mathcal{B}_1(\Gamma) \leq C \text{length}(\Gamma) + e \quad (3.18)$$

hold with probability at least $1 - \delta$. The constant C is independent of δ, e and the required mesh resolution; it is universal away from enlarged patches. Prescribed summable failure probabilities and local errors can be imposed simultaneously on a countable locally finite family of curves. In a deep anisotropic lattice patch there is also a grid-unit version: rescale to unit grid aspect, estimate length there, and convert the weight back using the longest physical side, with a constant independent of the aspect ratio in those units.

Proof. Discard an arbitrarily small amount of length with multiplicity and decompose the remainder into finitely many compact pieces on C^1 graphs of arbitrarily small slope s over straight axes. Such pieces may be taken injectively from the regular parameter branches, so multiplicity is accounted for by separate pieces. Cover their axis projections by finitely many intervals of total length at most the sum of their graph-piece lengths plus an arbitrarily small error. Extend the graphs over these intervals with comparably small slopes.

For each fixed relative-coordinate value z in the joint integration formula (3.10), classify a dual plane by its unit normal $\nu(z)$ and the graph axis e . If $|\nu \cdot e| > 2s$, the restriction of its affine defining function to the graph is strictly monotone. Each such plane patch meets the graph over its interval at most once. Charge a contributing tetrahedron to an axis interval of length comparable to d centered at a hit. Intersecting charged intervals have comparable scales, by slow grading, and their vertices belong to a bounded local candidate list. Thus the charges have bounded overlap. The sum of mesh diameters, and hence of the flat edge weights, is bounded by a constant times the graph length plus the local mesh upper bound at each interval endpoint. After the finite cover is fixed, all endpoint overhangs can be made as small as desired by the mesh choice.

For the remaining normals, the translation-coarea factor on this graph is at most Cs . For each fixed z , integration over the physical translation variable bounds the crossing integral by Csd^2 times the local graph length. Multiply by the joint density bound Cd^{-3} and the edge weight Cd , and then integrate over the bounded relative-coordinate set. This gives expected cost at most Cs times the graph length. The normal classification is translation invariant, so it is legitimate inside this unnormalized joint integral. The omitted trace length has correspondingly small expected cost by Lemma 3.16. Choose the omitted length and s sufficiently small, then choose the mesh bounds for the finite graph cover. Markov's inequality applies only to these small remainders, and makes their failure probability below δ without changing the constant on the retained graph length. This proves (3.18).

For a locally finite family assign summable failure probabilities and errors and impose the corresponding local upper bounds together. A union bound proves the simultaneous assertion. In a deep lattice patch perform the entire argument after the anisotropic rescaling. There are finitely many rational tie-plane families in these units; the same argument, or translated Riemann sums for their projected length coarea, gives the claimed constant. Physical flat weights are bounded by the longest-side factor times their grid-unit weights. $\square$

Lemma 3.18 (Sharp costs along a thin-grid subspace). *In a deep lattice patch adapted to a fixed k -plane E , use side d along E and side ηd along $E^\perp$, where $k \in \{1, 2\}$. For $m = k$, the*

translation-averaged tie-budget coefficient on a simple k -vector tangent to E is at most $1 + C\eta$ times its Euclidean norm. For general measured m -vectors, tangent contributions have constants uniform in η , and errors have a finite constant C_η . After a fixed linear output normalization A which keeps the two subspaces orthogonal, the sharp coefficient relative to the normalized tangent length or area is at most $1 + C_A\eta$. Here C_A depends only on $\|A\|_{\text{op}}$ and $\|A^{-1}\|_{\text{op}}$. The tangent bounds remain uniform in η , while the error constants may also depend on these two norms. The grid-unit bound of Lemma 3.17 also has no aspect-dependent factor on tangent lengths.

Proof. Rescale the grid to unit cubes, temporarily recording physical flat weights separately. For $k = 2$ slice a cube at a generic height in its thin direction. Each Kuhn tetrahedron projects to a triangle with one repeated vertex. Write the two unsplit projected barycentric masses as q_s, q_t . On a fixed-height slice, either part λ_0 of the split mass is a constant minus a sum of some of q_s, q_t , with coefficients zero or one. Consequently

$$\det D_{(q_s, q_t)}(q_s - \lambda_0, q_t - \lambda_0) > 0.$$

These are also the orientation coordinates for a leader triangle with three different projected vertices. Thus every such projected triangle is counted positively.

The total projected weight on a horizontal period is exactly its area. One way to see this without estimating individual positions is to take a constant large a in (3.12). Projection of this periodic simplexwise map on the horizontal slice is a torus map with periodic displacement and degree one. Its signed Jacobian integrates to one base area. Lemma 3.14 converts this integral, in the large- a limit, into the signed projected triangle weights. Generic heights give only finitely many transverse hits away from simplex boundaries; the signs just computed are positive, and repeated projections contribute zero. For a direct check, the two Kuhn tetrahedra whose vertical step comes first supply total projected area one at heights $2/3 < z < 1$, those whose vertical step is middle supply one for $1/3 < z < 2/3$, and those whose vertical step is last supply one for $0 < z < 1/3$. Each nondegenerate leader triangle contributes $1/2$ at its applicable height. The finitely many exceptional heights do not affect translation averaging.

In physical units a nondegenerate projected triangle differs from its projected area by at most $C\eta$ times the base area; a triangle with repeated projections has area at most $C\eta d^2$. The number of hits per period is uniformly bounded. The total physical area cost is therefore at most $(1 + C\eta)d^2$.

For $k = 1$, fix generic coordinates transverse to the long axis and traverse one axial period. Within a Kuhn tetrahedron the varying barycentric pair transfers mass from its lower axial position to its upper axial position, while the other masses are fixed. A switch between leaders with different axial projections is consequently always forward. Two fixed unsplit masses do not tie at a generic transverse coordinate. The projected periodic map has degree one, so total projected length is exactly the axial period. Edges with equal axial projection, and the transverse parts of the other edges, cost only $O(\eta)$ in period units. This proves the sharp length coefficient.

Translation coarea expresses each coefficient on a trace Jacobian as a finite homogeneous sum of absolute linear forms on its m -vector. The parallel-vector computations therefore give the asserted tangent coefficients. Splitting a vector into its tangent part and its error and using the triangle inequality gives a finite C_η on the error. A fixed normalization preserving orthogonality of the two subspaces changes these constants only through its two norm bounds. The final length assertion follows by making the anisotropic rescaling before applying Lemma 3.17. $\square$

Remark 3.19 (Order of choices for weighted budgets). Additional compact configurations and continuous weights may be fixed before selecting the target mesh bounds. They change how fine the mesh must be for the preceding estimates, but not the constants in those estimates. Small open

exemptions, their buffered no-movement cores, and strict angular safety conditions are included in this convention. Squeezing floors are selected only after the regular transverse map G and its compact measurement lists are fixed.

3.7 Source meshes with uniform shapes

Lemma 3.20 (Shape-controlled source sampling). *There are locally finite source meshes of uniform shape with arbitrarily small local upper size bounds, preserving finitely many separated deep cubical lattice patches in prescribed orientations and with prescribed common fine scales. Deep cells may remain cubes; elsewhere they are tetrahedra, with matching facet diagonals at interfaces. For an integrable nonnegative function g and $m \in \{1, 2\}$, their sampled m -faces satisfy the local estimate*

$$\mathbb{E} \sum_F \ell_F^{3-m} \int_F g d\mathcal{H}^m \leq C \int g, \quad (3.19)$$

with comparable local enlargements and bounded overlap understood. In particular this applies to $g = |Df|^p$ and controls the tangential derivative trace integrals. Lattice shifts can also be uniform over any prescribed macro-period formed by grouping fine cubes.

Proof. Use the construction of Lemma 3.12, now with cubical rather than anisotropic deep patches, and impose uniform altitude avoidance on the unstructured sites. Conditional on the lattice shifts, enumerate these sites and place them successively. When placing a site, exclude a neighborhood of thickness ε_0 times its scale around the affine span of every nearby tuple of at most three earlier or lattice vertices. Candidate neighbor lists are fixed and bounded by localization. A neighborhood of each such plane, line, or point removes at most $C\varepsilon_0$ of the jitter volume. For an absolute sufficiently small ε_0 , at least half the jitter volume remains. Choose uniformly in the remaining portion.

Any affinely independent lattice prefix of a candidate simplex comes from one rotated cubical lattice. There are only finitely many relevant normalized configurations, so its nonzero determinant or lower-dimensional volume has an absolute lower bound. Each subsequently placed vertex has altitude at least ε_0 times scale over the affine span of the preceding vertices. Induction gives a uniform lower bound on every nondegenerate simplex determinant after scaling. Together with the upper diameter bound this gives uniform shapes. Fully lattice simplices have the usual finite collection of Kuhn shapes. Deep cells may be left cubical, using the same diagonals on shared facets when meeting triangulated cells. Block modifications can consequently be kept inside cubical regions, with whole shared squares on their outer boundaries.

Given the previously placed sites and the lattice shifts, each new jitter has conditional density at most twice its original uniform density. Successive integration of the unselected sites, or the corresponding iterated-expectation bound on rectangular events, therefore bounds the joint density of any fixed tuple of k selected unstructured sites by $C^k d^{-3k}$, uniformly in the lattice shifts. The jitters need not be independent. Include the uniform density of a candidate's lattice shift, if present, and make the same change of variables as in Lemma 3.12. This again gives a joint density $q(t, z) \leq C d^{-3}$ for physical translation and normalized relative coordinates. Discard any later selection indicator. For each fixed z , an m -face has area $O(d^m)$, and Fubini gives

$$\int dt \int_{t+F_z} g d\mathcal{H}^m \leq C d^m \int_U g,$$

where U is the candidate's comparable enlarged neighborhood; only translations in its bounded support are included on the left. Multiplying by $C d^{-3}$, then integrating z over its bounded set,

gives $Cd^{m-3} \int_U g$ for the expected face integral. Its mesh scale is comparable to d by shape control. Multiplication by ℓ_F^{3-m} , followed by bounded candidate counts and overlap, proves (3.19).

For grouping fine cubes into macro-cubes, choose independently a uniform fine-period shift and a uniform discrete grouping offset. Their sum is uniform over the macro-period. Truncation of a lattice list concerns sites inside its fixed buffered patch; it does not translate the transition-site lists by many mesh steps. Thus every mixed local candidate still uses only one fine-scale physical translation variable after subtracting that shift and normalizing its jitter coordinates. Its bounded joint density, and hence the unnormalized joint Fubini calculation, remain unchanged. $\square$

4 Approximation for the low exponents

We prove the approximation theorem for $1 \leq p \leq 2$. The local constructions in this section use the curve and surface budgets of the preceding section. Surface budgets, and the inverse- BV theorem, are used only when $p = 2$.

Theorem 4.1 (Low-exponent approximation). *Let $\Omega, \Omega' \subset \mathbb{R}^3$ be bounded domains, let $1 \leq p \leq 2$, and let $f : \Omega \rightarrow \Omega'$ be a homeomorphism in $W^{1,p}(\Omega; \mathbb{R}^3)$. For every $\varepsilon > 0$ there is a locally bi-Lipschitz homeomorphism $H : \Omega \rightarrow \Omega'$ such that*

$$\int_{\Omega} (|H - f|^p + |DH - Df|^p) < \varepsilon. \quad (4.1)$$

In particular, the target of H is exactly Ω' , and $H \in W^{1,p}(\Omega; \mathbb{R}^3)$.

We work on finitely many compact sets carrying controlled jets. At rank three, radial blocks recover those jets; at ranks one and two, kernel compression reduces the core estimate to a line or plane section, with a further planar radial construction at rank two when $p < 2$. Boundary parametrization completes the remaining cells, and the final proof records the order of choices and proves global summability.

Throughout the section, all modifications are made inside the open domains. Constants denoted by C depend only on the dimension, p , and the fixed shape constants. A subscript records any additional dependence. In an expression $o(1)$ involving a mesh size, all the other parameters and the finite compact configuration have already been fixed. The quantities tending to zero will always be nonnegative costs or upper bounds for such costs.

4.1 Composition at a collapsed stratum

We first record the regularity fact used when a three-dimensional region is compressed toward a line or a plane. It also explains why the finite-piece regularity in the budget construction is retained.

Lemma 4.2 (Tangential composition limit). *Let D be a compact Lipschitz parameter piece of dimension d , and let $q_j, q_0 : D \rightarrow U$ be Lipschitz maps into an open subset of a Euclidean space. Suppose that their images lie in one compact subset of U , that $q_j \rightarrow q_0$ uniformly, and that*

$$Dq_j \longrightarrow Dq_0 \quad \text{a.e. on } D, \quad \sup_j \|Dq_j\|_{L^\infty(D)} < \infty.$$

Let $v : U \rightarrow \mathbb{R}^a$ be Lipschitz on a neighborhood of that compact set. Suppose there are finitely many closed pieces C_ν covering the neighborhood and C^1 maps v_ν , defined on neighborhoods of C_ν , such that $v = v_\nu$ on C_ν . Then, for every finite $r \geq 1$,

$$D(v \circ q_j) \longrightarrow D(v \circ q_0) \quad \text{in } L^r(D). \quad (4.2)$$

No rank assumption is imposed on Dq_0 . The assertion applies separately to an edge, a face, or a volume piece.

Proof. For each $j \geq 0$ and each ν , apply the density theorem on the parameter piece to $q_j^{-1}(C_\nu)$. At almost every point of this preimage, locality of the weak derivative gives

$$D(v \circ q_j) = Dv_\nu(q_j)Dq_j.$$

Remove the union of the exceptional sets for the countably many j and finitely many ν . At a remaining parameter point, every branch containing q_0 gives the same product with Dq_0 , namely $D(v \circ q_0)$. If a branch occurs along infinitely many q_j , its closedness implies that it contains q_0 . Continuity of its derivative and convergence of Dq_j therefore show that its products converge to this common value. Finiteness of the branch list proves pointwise convergence of the full sequence. The fixed derivative bounds give a bounded majorant, so dominated convergence proves (4.2). The proof takes place on D ; it does not pull back an ambient null set through a possibly rank-deficient map. $\square$

4.2 Full-column-rank radial blocks

Here the parameter dimension is either $m = 3$, with $1 \leq p \leq 2$, or $m = 2$, with $1 \leq p < 2$. In the latter case the parameter domain is a flat plane section of the source. All target motions remain ambient three-dimensional homeomorphisms.

Let K be a compact subset of such a parameter patch. Assume that there is a local C^1 ambient diffeomorphism b such that

$$f = b, \quad D_m f = D_m b \quad \text{on } K \quad (4.3)$$

up to the null sets of the indicated traces. The map f on the patch is Sobolev and has its continuous ambient values. Both Db and $(Db)^{-1}$ are bounded by a fixed constant, denoted collectively by C_b . For $m = 2$, the additional normal column of b is chosen so that its ambient orientation agrees with that of f . Write

$$F = b^{-1} \circ f.$$

Thus $F = \text{id}$ and $D_m F = I_m$ on K , where I_m is the derivative of the identity inclusion of the parameter plane into $\mathbb{R}^3$. Target coordinates in the following construction are always the b -coordinates. If $m = 3$ and $p = 2$, then

$$F^{-1} \in BV_{\text{loc}}, \quad (4.4)$$

by [14, Theorem 1.1]. That theorem requires no finite-distortion hypothesis.

Choose an integer N , put $l = N^{-1}$, and fix a sufficiently large absolute constant K_0 . Set

$$\gamma = l/K_0. \quad (4.5)$$

Use a uniformly shifted cubical grid of side h , with each macro cube subdivided into tiles of side lh . The word “cube” includes a square when $m = 2$. Cubes meeting K are candidates. In each candidate Q , draw a center a uniformly in a central cube of side $h/4$; retain only centers belonging to K . Rays from a through boundary mesh vertices are the forward spokes. When $m = 3, p = 2$, also choose, independently on each boundary mesh square, a point u_j uniformly in its central half and use the target ray from a through u_j .

Cubical polar coordinates are written

$$x = a + r(\omega - a), \quad \omega \in \partial Q,$$

so the boundary has radial coordinate $r = 1$. In dimension two, the ambient output box is the normal thickening of Q , centered on the parameter plane and of height h .

Lemma 4.3 (Radial block construction). *Fix finitely many separated compact configurations satisfying (4.3), and fix l . For sufficiently small h one can retain candidate blocks with the following properties.*

1. *The expected m -volume of discarded candidate blocks tends to zero as $h \rightarrow 0$. The union of retained blocks therefore differs from K by a set whose expected measure tends to zero.*
2. *Each retained Q has the core*

$$B_Q = a + (1 - 8\gamma)(Q - a).$$

The complement consists of shape-controlled frusta of scale lh , one over each boundary tile. After the common budget construction and the source and target changes described below, the map on B_Q is b , except on caps of total measure at most $C\gamma|Q|$. Its tangential derivative on the whole core and its core boundary is bounded by C_b .

3. *Every radial connector in the final shell has image length at most $C_b\gamma h$. Nonincident mesh edges avoid the radial-angular expansions. Subsequent cap contractions multiply their length budgets by at most a fixed constant. With $\ell = lh$, the converted shell edge cost after constant-speed parametrization is $\ell^{m-p} \sum_e L_e^p$, where L_e is the image length. Its bound is $O_b(l) \sum_Q |Q| + o(1)$ after the source selection and individual curve-budget bounds, with arbitrary additional transfer errors.*
4. *If $m = 3, p = 2$, include the full wedge from a to each boundary tile edge among the measured surfaces. There are bounded continuous weights, fixed before the target budget mesh, for which the final connector-face areas are bounded by the weighted H_0 -areas, up to arbitrarily small errors. After multiplication by lh , the expectation of their total converted budget $\mathcal{B}_{\text{wedge}}$ satisfies*

$$\mathbb{E}\mathcal{B}_{\text{wedge}} \leq C_b\gamma \sum_i |K_i| + o(1). \quad (4.6)$$

Outer ordinary edges and faces have converted cost $O_b(l) \sum_Q |Q| + o(1)$ on these configurations.

The construction can be performed simultaneously with the other budgeted pieces used below. In a volume block the only measured interior surfaces are its wedges. In the plane-section application, any additional graph to be fixed is assumed to have positive clearance from K and its image. Taking h smaller then keeps all expansion and cap supports off that graph. More precisely, write V for the ambient initial map after its output motions and $q_Q : Q \rightarrow Q$ for the radial parameter change. The controlled trace is $V \circ q_Q$. For $m = 3$ these parameter changes are ambient source self-homeomorphisms; for $m = 2$ they are retained solely as section parametrizations. The graph preservation means that the radial motions leave its already-budgeted H_0 -image unchanged; it does not assert equality of H_0 with f on that graph.

Proof. We give the selection, the maps, and the estimates in their order of use.

Selection of centers and stars. Averaging the macro shift and the center draw is equivalent, up to fixed bounded densities, to integration in the actual center a and normalized relative offset variables. The weight in a block sum is h^m . Failure to choose $a \in K$ on candidate blocks costs at most

$$C|N_{Ch}(K) \setminus K|,$$

which tends to zero. Here $N_t(K)$ denotes the parameter-space t -neighborhood of K .

For almost every center in K and almost every fixed choice of relative variables, each of the finitely many forward line traces is absolutely continuous and has derivative the identity at a in the one-dimensional Lebesgue-point sense. This follows from slicing and $D_m F = I_m$ on K . In the reverse system, use (4.4) and $F^{-1} = \text{id}$ on K , together with directional BV slicing and its separate absolutely continuous and singular variation identities [3, Section 3.11, Theorems 3.103, 3.107, and 3.108]. At almost every line density point of K , the absolutely continuous derivative of the inverse trace is the identity and the density of its singular variation is zero. The trace is continuous because F^{-1} is continuous. Consequently both systems have relative conical differentiability at a , and the length of an initial ray interval is its radius times $1 + o(1)$.

For completeness, near-minimal length gives the single-crossing sections required by Proposition 3.9. The radial distance along one distorted initial ray attains every level up to $(1 - o(1))t$ and has total variation at most $(1 + o(1))t$. One-dimensional coarea shows that levels with more than one crossing have measure $o(t)$. A finite union of these exceptional sets still has measure $o(t)$, so there are arbitrarily small common levels crossed once by every ray in the system. Injectivity and the local differentiability estimate exclude remote tails from these shrinking sections.

The loop condition in Proposition 3.9 is also obtained by slicing. Fix one of a countable library of finite piecewise smooth loop systems on the angular sphere. For a fixed library loop ω , set

$$E_t(a) = \int (|D_m F(a + t\omega(s)) - I_m|^p + \mathbf{1}_{\text{patch} \setminus K}(a + t\omega(s)) |\omega'(s)|) ds.$$

Extend the integrands to a fixed neighboring patch. Translation continuity and their vanishing on K give $\int_K E_t(a) da \rightarrow 0$. A diagonal choice of $t_n \downarrow 0$ with summable means for the first n library loops gives $E_{t_n}(a) \rightarrow 0$ for every loop at almost every center. Each loop then has a point in K , where $F = \text{id}$. Absolute continuity from that point gives uniform convergence of $(F(a + t_n\omega) - a)/t_n$ to ω , also when $p = 1$. The ray estimates hold at all sufficiently small scales, so this subsequence supplies the loop tests simultaneously with them. Choose the library fine enough to avoid the separated cones and test the required punctured-sphere generators. In the planar case equatorial circles suffice. Countability permits all these tests at the same generic centers.

For any prescribed small cone and positional tolerances depending on l , dominated convergence in the center and relative variables now shows that the failed tests have total block volume tending to zero. In full dimension the oriented generator test itself forces the comparison link to have the identity orientation; no separate Jacobian-sign assertion is needed here. In the planar case the orientation was fixed by the normal column of b .

Excluding intrusive edges. Call the unmodified fine lattice skeleton the default skeleton, whether or not a candidate block will later be replaced. Any default edge whose image can enter a proposed inset target support lies in a shrinking source neighborhood $U_h = N_{\rho(h)}(K)$ of K , by continuity of the inverse, where $\rho(h) \rightarrow 0$. Enlarge $\rho(h)$ by $O(h)$ to include each entire relevant edge. No estimate $\rho(h) = O(h)$ is needed or assumed. For an edge e meeting K , absolute continuity from an unchanged point gives

$$\sup_e |F - \text{id}| > c\gamma h \implies \int_e |D_t(F - \text{id})|^p \geq c(l, \gamma, p)h.$$

Translation averaging makes the sum of these integrals in the shrinking neighborhood $o(h^{1-m})$ for fixed l . Edges missing K are charged by their entire length lh to $U_h \setminus K$. Thus the expected number of bad edges is $o(h^{-m})$. Their total image length is $o(h^{1-m})$: use the same derivative-error estimate

and Hölder on the bad edges, and use the original derivative integral on those wholly off K . For $p = 1$ this is the direct length integral, with no Hölder gain required.

A rectifiable curve of length L meets at most $C(1 + L/h)$ grid blocks at scale h , by division into subarcs of length at most h . In the planar case first project the curve to the parameter plane in target coordinates. Discarding all blocks struck by bad-edge images therefore loses $o(1)$ total volume. Good nonincident edges are $o(\gamma h)$ from their original positions and avoid the strictly inset expansion supports. Accurate spoke positions also prevent a spoke of one retained block from entering another block's support.

The protected germ and the radial maps. Apply Proposition 3.9 at the retained centers. In the subsequent use of Lemma 3.10 and Proposition 3.15, require the initial map H_0 to satisfy

$$b^{-1} \circ H_0 = \text{id} \quad \text{near every retained center.}$$

Keep the narrow forward cones and the small positional errors. In the critical case, keep also the condition that the preimage of each chosen target ray, up to $r \leq 1 - \gamma$, lies in the corresponding open radial sector inside Q , except at a . The star modification preserves the angular conditions near the center. Away from a smaller center neighborhood the constraints have compact positive margins, and fine approximation with reciprocal control preserves them. The simplex displacement is turned off still nearer the germ. Thus the later budget transfer does not destroy the ray clearance. For now fix the germ neighborhoods and margins. The actual budget transfer is made only after the outer weight below has been fixed; the next paragraphs specify the maps that will then follow it.

Once H_0 is chosen, choose $\lambda > 0$ so small that $a + \lambda(Q - a)$ is in the exact germ. Precompose on Q with the radial homeomorphism fixing ∂Q and sending B_Q linearly onto this tiny cube. On the remaining ray segments use the linear radial interpolation. In target b -coordinates postcompose with a radial expansion $r \mapsto e(r)$ which is the identity for $r \geq 1 - 2\gamma$ and satisfies

$$e(r) = \frac{1 - 8\gamma}{\lambda} r \quad (0 \leq r \leq \lambda).$$

This gives exactly b on B_Q . For $p < 2$, any strictly increasing piecewise linear profile between these prescribed portions is sufficient. The supports for different blocks are disjoint.

Angular attenuation when $p = 2$. For this paragraph $m = 3$. On a boundary square S_j , write its points in rays from u_j as

$$\omega = u_j + \rho R_j(\theta) v(\theta), \quad 0 \leq \rho \leq 1.$$

Here $R_j(\theta)$ is the distance to the square boundary. It is comparable to lh and is piecewise smooth with the corresponding uniform derivative bounds. Choose $0 < \kappa, t < 1/4$ and replace ρ by

$$\phi(\rho) = \begin{cases} (1 - t)\rho/\kappa, & 0 \leq \rho \leq \kappa, \\ 1 - t + t(\rho - \kappa)/(1 - \kappa), & \kappa \leq \rho \leq 1. \end{cases}$$

The resulting map A preserves S_j , fixes its boundary, and expands its tiny central copy. On each smooth angular piece,

$$\det DA = \phi'(\rho) \frac{\phi(\rho)}{\rho}, \quad |DA| \leq C \left(1 + \frac{lh}{|\omega - u_j|} \right). \quad (4.7)$$

The angular dependence of R_j adds shear but leaves the displayed determinant unchanged. On a compact set missing u_j , choose κ below its relative distance and then let $t \downarrow 0$; the determinant tends uniformly to zero there. For the straight interpolation $A_\beta = (1 - \beta) \text{id} + \beta A$, the radial profile is $(1 - \beta)\rho + \beta\phi(\rho)$. Away from the central copy its radial slope is at most one, while its transverse expansion has the bound in (4.7). Hence

$$|\det DA_\beta| \leq C \left(1 + \frac{lh}{|\omega - u_j|} \right), \quad 0 \leq \beta \leq 1, \quad \rho \geq \kappa. \quad (4.8)$$

Choose $0 < \lambda < \lambda' < 1/4$ still inside the exact affine region. Turn A on between λ and λ' , leave it on through $1 - 4\gamma$, and turn it off between $1 - 4\gamma$ and $1 - 3\gamma$. Choose a slope $0 < \alpha < \gamma$ on $[\lambda, 1/2]$, and put

$$k_\alpha = \frac{6\gamma - \alpha(1/2 - \lambda)}{1/2 - 2\gamma} > 0.$$

The denominator is bounded below, so $k_\alpha \leq C\gamma$. On $[1/2, 1 - 2\gamma]$ use this slope. Explicitly, these portions are

$$e(r) = \begin{cases} 1 - 8\gamma + \alpha(r - \lambda), & \lambda \leq r \leq 1/2, \\ 1 - 8\gamma + \alpha(1/2 - \lambda) + k_\alpha(r - 1/2), & 1/2 \leq r \leq 1 - 2\gamma. \end{cases}$$

They meet $e(1 - 2\gamma) = 1 - 2\gamma$ exactly. The slope α can still be decreased as required by the inner error estimate. All maps for fixed positive parameters are bi-Lipschitz with finitely many closed C^1 pieces.

In normalized cubical polar coordinates the target map is

$$(r, \omega) \mapsto a + e(r)(A_{\beta(r)}(\omega) - a).$$

Its two-angular-direction area factor is bounded by

$$C_b(e/r)^2 |\det DA_{\beta(r)}|, \quad (4.9)$$

and its mixed radial-angular factor is bounded by

$$C_b(e/r)(|e'| + e|\beta'|)|DA_{\beta(r)}|. \quad (4.10)$$

Indeed the radial derivative is $e'(A_\beta - a) + e\beta'(A - \text{id})$, the angular derivative is eDA_β , and $|A - \text{id}| \leq Clh$. Uniform transversality of the cubical radial direction to a boundary tile gives these estimates for arbitrary tangent two-planes as well.

These two estimates treat different components of a surface tangent plane. Its purely angular component is controlled by the determinant of $DA_{\beta(r)}$, while its mixed component uses only the first power of $|DA_{\beta(r)}|$. Write

$$G(\omega) = 1 + \frac{lh}{|\omega - u_j|}$$

on each boundary tile. Figure 2 displays the resulting area bounds in the unexpanded target coordinates. The inner costs will be made small after H_0 has been constructed, using its actual measured surfaces. The remaining costs will be bounded by a continuous outer weight fixed before the budget transfer. This is why the later attenuation choices do not change the budget already selected.

The measured connector surfaces after source squeezing are truncated wedges. Between λ and λ' they are exactly radial over tile edges and are unchanged by A ; their cost is arbitrarily small

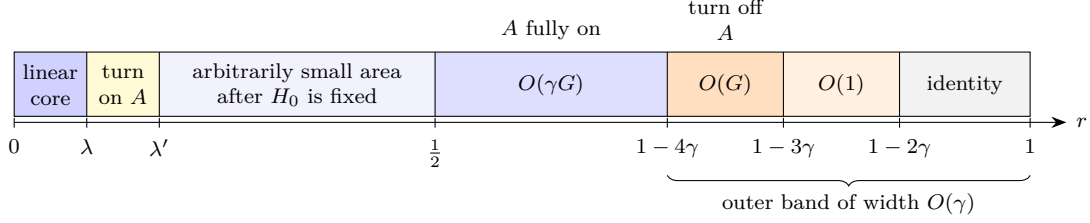


Figure 2: Critical radial area costs in the pre-expansion coordinate r ; intervals are not drawn to scale. Here $G = 1 + lh / \text{dist}(\omega, u_j)$ has bounded angular mean. The linear core contains no uncontrolled connector face. Inner costs can be suppressed after H_0 is fixed. Among the remaining terms, the middle band carries the factor γ , and terms without this factor occur only in the thin outer band.

with the increase of e on that interval. Below λ these surfaces are absent. On $[\lambda', 1/2]$, all relevant surfaces have positive angular clearance from the selected rays. After H_0 and λ' are fixed, choosing κ, t sufficiently small suppresses (4.9), and choosing e' sufficiently small suppresses (4.10). Only finiteness of the fixed surface areas is used in this step. More explicitly, work in units $h = 1$ and let

$$M_0 = \int_{\lambda' \leq r \leq 1-4\gamma} G(\omega) d\mu < \infty,$$

where μ is the unexpanded H_0 -area of the finitely many surfaces. Choose κ below the angular clearance of every measured surface on $\lambda' \leq r \leq 1 - \gamma$, including the switch-off band. This clearance follows from the inverse-ray sector condition and compactness after removal of the exact germ. Thus every use of (4.8) on these surfaces lies in $\rho \geq \kappa$. The suppressed error is bounded by

$$C_b \left(\frac{t}{(\lambda')^2} + \frac{\alpha}{\lambda'} \right) M_0.$$

The exact radial wedges on $[\lambda, \lambda']$ cost at most $C_b L \alpha (\lambda' - \lambda)$, where L is their total angular edge length. Thus t and α can make the sum smaller than any given positive error, while k_α remains positive and bounded by $C\gamma$.

On the remaining fully-on band the mixed cost is at most $C_b \gamma (1 + lh/|\omega - u_j|)$ times unexpanded area; the two-angular cost can again be suppressed. On the switch-off band, $l|\beta'| \leq Cl/\gamma = CK_0$, so (4.8)–(4.10) bound both costs by a constant times the *first* power of $1 + lh/|\omega - u_j|$. The radial-only outer end has a fixed bounded area factor. Squaring the angular factor would not give the needed averaging estimate, and is avoided by the determinant calculation.

A weight fixed before budgeting. Inside a retained volume block, the measured surface list consists only of its full wedges. Every other measured source surface lies outside the block interior; kernel templates remain in their own macro blocks. The preimage of a selected target ray lies in the open sector over its tile, so it meets neither these outside surfaces nor a wedge, which lies on a sector boundary. On compact ray segments away from the exact germ, local finiteness therefore gives positive clearance from the entire measured surface list.

In the outer radial region choose a continuous buffered majorant W . The angular envelope on a tile is a buffered version of

$$1 + \frac{lh}{|\omega - u_j|}.$$

Its singularity can be capped in a tiny neighborhood of the ray: the ray-preimage sector condition makes the outer compact portions of all measured surfaces disjoint from that ray. First choose the

caps for the adjusted topological map with a positive margin, then choose the budget mesh and G, T fine enough to preserve the margin. Cutoffs in disks of radius $O(lh)$ and a constant background make the envelope continuous across tile boundaries. Write the resulting bounded envelope as $\widehat{G}$. It agrees with an upper bound for the uncapped factor on the measured surfaces and is bounded above by a fixed multiple of the uncapped integrable envelope everywhere.

Choose continuous radial cutoffs with the following values, using linear interpolation between the listed bands:

	0	1
χ_{low}	$r \leq 1/2 - \gamma$	$r \geq 1/2$
χ_{out}	$r \leq 1 - 6\gamma$	$r \geq 1 - 5\gamma$
χ_{cut}	$r \geq 1 - \gamma$	$r \leq 1 - 2\gamma$.

For a sufficiently large fixed C_b , take

$$W = \chi_{\text{out}} + C_b \chi_{\text{low}} \chi_{\text{cut}} (\gamma + \chi_{\text{out}}) \widehat{G}. \quad (4.11)$$

It is zero for $r \leq 1/2 - \gamma$, dominates the required $C_b \gamma G$ on the middle band and $C_b G$ on the return ramp, and equals the ordinary weight one for $r \geq 1 - \gamma$. The radial outer end is covered as well, and the angular factors are confined below $1 - \gamma$. The part without the factor γ occupies an $O(\gamma)$ radial band. Buffered cutoffs may enlarge these bands by another fixed multiple of γ without changing the estimates.

This fixes a bounded continuous weight before Proposition 3.15 is applied. The weight vanishes in the star-modification neighborhoods, and the simplex map is the identity on smaller exact-germ neighborhoods. Thus the measured crossings used by the transfer are the original f -crossings. The mesh-element suprema of W give the weighted H_0 -area upper bound. The still smaller attenuation parameters in the preceding paragraph are chosen *after* H_0 ; they need not have been known when W was fixed. Equations (4.9) and (4.10) prove the asserted weighted bound for all connector, outer, or nonincident measured surfaces, with arbitrarily small additive errors.

Spokes and the final contractions. After the expansion, each shell spoke lies in a ball of radius $C\gamma h$ about its intended outer mesh vertex. Its entire image is outside the open linear λ -core by injectivity and the exact germ, with the inner endpoint on that core's boundary. Its angular direction remains in a sufficiently narrow vertex cone, and along the whole trace its pre-expansion radius is at most $1 + o(\gamma)$. The angular maps fix vertices and have uniform local Lipschitz bounds near them. These facts give the stated ball inclusion.

Each such spoke has finite length. Homothetically contract its entire vertex ball to a sufficiently tiny concentric ball, and extend radially to the identity on the twice-larger ball. The extension has a uniform upper Lipschitz bound, while the homothety factor may be as small as required to reduce the whole spoke length to $C_b \gamma h$. Group all spokes at a shared vertex and use the same contraction for them. The twice-larger balls are pairwise disjoint when K_0 is large enough. Their centers are outside the open cores. There are $O(l^{1-m})$ boundary vertices, so their intersection with a core has measure at most

$$Cl^{1-m}(\gamma h)^m \leq C\gamma h^m.$$

On the core the map before contraction is b ; hence the contracted map still has derivative bounded by C_b . All other length and area estimates are multiplied by a bounded factor. No uniform inverse Lipschitz bound in the contraction factor is required.

Averaging the wedge weight. The total area of the full wedges in a three-dimensional block is $O(h^2/l)$. On their portions in K , $F = \text{id}$. The outer band has relative radial thickness $O(\gamma)$, and the remaining nonzero weight has the factor γ . Also, for a point ω fixed before the choice of u_j ,

$$\mathbb{E}_{u_j} \left[1 + \frac{lh}{|\omega - u_j|} \right] \leq C,$$

because u_j is sampled in two dimensions at scale lh . An indicator of successful selection can be dropped in this upper bound. Multiplying the resulting weighted area by lh gives $C_b \gamma h^3$ per block.

On off- K wedge portions, use the same uncapped angular envelope before making any clearance-dependent choices. For normalized wedge parameters v , translation continuity gives

$$\int_K (|Df|^2 \mathbf{1}_{\Omega \setminus K})(a + hv) da \rightarrow 0,$$

and integration in the relative variables and wedge parameters gives the corresponding averaged trace estimate. The generic surface formula in Lemma 3.11 applies. This proves (4.6), including the later target-mesh averaging of Lemma 3.16. The outer grid has only $O(l^{1-m})$ edges or faces at scale lh per macro boundary, so its converted bounded-derivative cost is $O_b(l)|Q|$. Its off- K error tends to zero by the same trace averaging. Nonincident edges avoid the expansion; nonincident surfaces use the same clearance and weighted estimate. All required assertions follow. $\square$

4.3 Compact jets and kernel blocks

We next reduce the remaining positive-rank sets to line and plane sections. The distinction between invertible, zero, and nonzero singular derivatives already plays a central role in the planar construction of Hencl–Pratelli [24, Section 1.1 and Section 3]. Here the singular models are handled by compressing their kernel directions. A matrix with a prescribed rank need not be the derivative of any ambient diffeomorphism; its kernel is used only to choose the source coordinates.

Lemma 4.4 (Finite compact jet selection). *Given $\varepsilon_0 > 0$, one can choose $M < \infty$ and finitely many pairwise disjoint compact sets $K_i \subseteq \Omega$, of ranks $k_i \in \{1, 2, 3\}$, such that*

$$\int_{\Omega \setminus \bigcup_i K_i} |Df|^p < \varepsilon_0. \quad (4.12)$$

On their nonzero singular values there are common bounds M^{-1} and M . At rank three the sets have ambient C^1 diffeomorphism charts b_i agreeing with the values and derivatives of f , with chart bounds C_M . At ranks one and two, for any subsequently chosen $\delta > 0$, the pieces can be refined so that

$$|Df - D_i| \leq \delta \quad \text{on } K_i, \quad \text{rank } D_i = k_i, \quad (4.13)$$

with the same type of singular-value bounds. The compact pieces can be enclosed in disjoint source neighborhoods and corresponding disjoint target neighborhoods with buffers. Once all finite multiplicative constants are fixed, the neighborhoods can be chosen so that

$$\sum_i \int_{U_i \setminus K_i} (1 + |Df|^p) \quad (4.14)$$

is as small as prescribed, and the inverse images of the buffered target neighborhoods lie in the corresponding U_i .

Proof. Sobolev maps are approximately differentiable almost everywhere. Truncate the positive-rank sets where their nonzero singular values lie in $[M^{-1}, M]$, and choose M so large that the lost derivative integral is small. Rank-zero points have zero derivative energy and need not be included. On each retained part, pass to compact subsets with continuous jet data and uniform approximate differentiability, losing an arbitrarily small further derivative integral.

These are C^1 Whitney jets after a further compact refinement. To see the value condition, take two nearby retained points and compare their approximate first-order expansions at a common good point in the overlap of balls of radius comparable to their separation. The uniform density losses can be made smaller than a fixed fraction of the overlap, so such a point exists. Subtracting the two expansions, using continuity of the jets, and then sending the differentiation error to zero gives the Whitney remainder estimate. The C^1 extension theorem [41, Part I, Section 4, Theorem I] supplies the extensions. At full rank, continuity of the derivative and the inverse function theorem give local invertibility. Injectivity on the compact set, together with compactness, makes the extension injective on a sufficiently small neighborhood of the whole compact piece. Its derivative and inverse derivative bounds can be kept within C_M .

For ranks one and two, cover the bounded rank- k matrix range by finitely many δ -balls centered at rank- k matrices with comparable nonzero singular values. Refine the compact pieces accordingly, losing an arbitrarily small integral to make the finite pieces disjoint. Their number does not enter the local chart bounds. Injectivity of f makes their compact images disjoint as well. Regularity of the integrable measure $(1 + |Df|^p) dx$ gives (4.14); continuity of f^{-1} permits target buffers whose preimages stay inside these source neighborhoods. The initial losses can be allocated to give (4.12). $\square$

Use Lemma 3.20 with mesh sizes at most lh , deep cubical regions around the K_i , and a common macro shift within each such region. At rank $k = 1, 2$, choose orthonormal coordinates $x = (y, z)$ with $z \in \ker D_i$ and $y \in \mathbb{R}^k$. At rank three use Lemma 4.3. A failed macro block is left as ordinary fine cells. Outside the retained macro blocks all cells are ordinary, including the locally finite cells approaching $\partial\Omega$.

The anisotropic target grids of Lemma 3.18 are needed only at rank one and at rank two when $p = 2$. Their long directions span range D_i , their aspect parameter is $\eta > 0$, and a linear normalization L_i can be chosen so that

$$L_i D_i(y, z) = (y, 0), \quad \|L_i\| + \|L_i^{-1}\| \leq C_M. \quad (4.15)$$

The range and its orthogonal complement remain orthogonal in this normalization. Elsewhere the target grid can be isotropic.

Consider one rank- k candidate macro cube Q of side h . Let B be its concentric product cube obtained by removing one fine layer from each side, let Y be its projection onto the y -coordinates, and let z_0 be its central kernel coordinate. Put

$$S = Y \times \{z_0\}.$$

Construct a Lipschitz map

$$P_0(y, z) = (y, p_y(z)) : Q \longrightarrow Q \quad (4.16)$$

which is the identity on ∂Q and collapses B onto S . For $y \in Y$, collapse the inner kernel cube to z_0 and extend radially and linearly to the outer kernel boundary. Outside Y , blend with the identity using a piecewise smooth y -cutoff. These formulas may be chosen semialgebraic and piecewise C^1 on closed pieces, with Lipschitz bound C_l . They preserve y exactly, so

$$D_i D P_0 = D_i. \quad (4.17)$$

For $0 < \lambda \leq 1$, put

$$P_\lambda = (1 - \lambda)P_0 + \lambda \text{id}. \quad (4.18)$$

For each fixed y the kernel radial slopes are positive. The map is triangular in (y, z) , fixes the boundary, and is bi-Lipschitz on Q .

Budget the shell edges after P_0 , and also its shell faces when $p = 2$. Include S , its tile edges, and all lower-dimensional intersections that occur in these templates. In source and target units divided by h , require that the sum of the p -power tangential derivative errors from the model on these finitely many pieces is at most τ^p . On a P_0 -image this means the error from $D_i DP_0 = D_i$; on S it means the error from its y -columns. The unprojected section is required to carry its Sobolev trace even when $k = 2, p < 2$.

Lemma 4.5 (Successful kernel blocks). *For fixed M, l, τ , the total expected volume of candidate rank-one or rank-two blocks failing these trace tests tends to zero as first $\delta \rightarrow 0$ and then $h \rightarrow 0$, with sufficiently small excess neighborhoods. In successful blocks the transferred unparametrized shell data have converted cost*

$$O_M(l) \sum_Q |Q| + \text{an arbitrarily small error}. \quad (4.19)$$

Here converted edge cost means $(lh)^{3-p}$ times image length to the power p , and, when $p = 2$, converted face cost means lh times image area. Fixed factors depending on l and η are allowed on the trace errors.

Proof. Translation averaging on the fixed product templates, with block weight h^3 , bounds the expected sum of normalized trace errors by

$$C_l \sum_i \int_{N_{Ch}(K_i)} |Df - D_i|^p. \quad (4.20)$$

The measured curve strata, and the measured surface strata when $p = 2$, are covered by Lemma 3.11. For the additional unprojected rank-two plane section when $p < 2$, ordinary Sobolev slicing and translation Fubini supply the $W^{1,p}$ trace used in these tests. This step requires neither an image-area formula nor a two-dimensional Lusin property. On K_i , the integrand is at most δ^p ; off K_i it is bounded by $C_M(1 + |Df|^p)$, whose excess integral is small. Markov's inequality therefore makes the failed volume small relative to the fixed threshold τ^p .

There are $O(l^{-2})$ shell cells of side lh . In grid units, with the longest side restored after the anisotropic rescaling, (4.17) prevents inflation of the model term. The factors from DP_0 and from anisotropy multiply only $Df - D_i$. Lemma 3.17, followed by Hölder on each curve, gives the edge length-power budgets. At $p = 2$, Lemma 3.16 and the squared derivative trace integrals give the face budgets. A bounded model derivative costs $C_M(lh)^3$ per shell cell after conversion; multiplying by $O(l^{-2})$ gives $C_M lh^3$ per macro block. For fixed l, η , the error contributions are made as small as desired by τ and the budget-transfer tolerances. All relevant images lie deep inside their prescribed target patches for small h , by the source and target buffers. This proves (4.19). $\square$

4.4 Parametrizing the retained line and plane sections

For a successful rank- k block we shall find a bi-Lipschitz self-homeomorphism σ_Y of Y , preserving its tile complex, and a tangential map h_Y into the target. If V denotes the initial map H_0 after all the required output motions, the relation is

$$h_Y(y) = V(\sigma_Y(y), z_0). \quad (4.21)$$

The aim is

$$\sum_{\text{successful } Q} h^{3-k} \int_Y |Dh_Y - D_i|_Y^p \quad \text{arbitrarily small.} \quad (4.22)$$

Only the y -columns of D_i occur in this formula. The factor h^{3-k} is the volume of the kernel fibers, up to fixed constants.

Rank one. On each interval tile, reparametrize the transferred curve at constant speed in the normalization L_i . By Lemma 3.18, its expected length budget, divided by the tile length, is at most

$$1 + C_M \eta + o_\tau(1) + o_{\text{transfer}}(1), \quad (4.23)$$

where l and η have already been fixed. Its budget is also at least $1 - o(1)$: the section trace test gives almost correct endpoint differences, and a sufficiently fine transfer preserves the values to a still smaller error. Consequently the total positive excess, weighted by tile length times h^2 , has arbitrarily small expectation.

The constant speeds are bounded by C_M simultaneously, using Lemma 3.17. To see that its grid-unit constants do not deteriorate with η , split the normalized trace derivative into the tangent model and the error. The model has size bounded by C_M in longest-side-restored grid units. The error has p -mean at most $C_{\eta,l}\tau$, and is made small after the grid parameters are fixed.

Here is the elementary sharpness calculation. On a unit normalized interval, write u for the constant-speed curve, $s = |u'|$, and $d = u(1) - u(0)$. If e is the desired unit direction, then

$$\int_0^1 |u' - e|^2 = s^2 + 1 - 2e \cdot d. \quad (4.24)$$

Since $s \leq C_M$, $|d - e|$ is small, and $s \geq 1 - o(1)$, the right side is bounded by $C_M(s - 1)_+$ plus an arbitrarily small error. For $1 \leq p \leq 2$, Hölder converts this squared error to a p -error. Rescaling the tiles and summing with the fiber weights proves (4.22). In particular the argument works at $p = 1$; it does not appeal to strict convexity of the L^1 norm.

Rank two at the critical exponent. Now $k = p = 2$. For every planar tile, use the area budget in the normalization L_i . Lemma 3.18, applied to its oriented tangent 2-vector and the squared derivative trace error, gives expected normalized area at most

$$1 + C_M \eta + o_\tau(1) + o_{\text{transfer}}(1). \quad (4.25)$$

There is also the lower bound $1 - o(1)$ for every sufficiently fine generic sample. Indeed the edge tests make the original boundary trace uniformly close, after translation and normalization, to the boundary of the identity square. The transfer preserves that closeness. Orthogonal projection of the transferred disk to the model plane has degree one at every point a small distance inside the square. It therefore covers that inner square, and its area is at least the inner-square area. Since the transferred area is no larger than its budget plus a chosen small error, the same lower bound holds for the budget. These tiles lie away from all exempt star centers.

It follows that the weighted sum of the positive area excesses has arbitrarily small expectation. To obtain the needed boundary parametrization, subdivide each common edge into J fixed intervals and use constant speed on each interval, preserving its endpoints. Choose J large before the finer trace and transfer tolerances. Lemma 3.17 bounds the normalized speeds by C_M ; the error part is $C_{\eta,l,J}\tau$ and is made small later. Every reparametrized point stays within its J -interval, so the

boundary values converge uniformly to the ideal translated square as $J \rightarrow \infty$ and the finer errors tend to zero. The boundary squared derivative integral is uniformly bounded.

Absorb these common edge changes by coning in each tile, then apply Lemma 3.3. Fix a desired derivative tolerance $\zeta > 0$. The sharp assertion of Lemma 3.3 supplies an area-excess tolerance $\rho > 0$ and a boundary-closeness tolerance, using the already fixed speed bound. On tiles whose normalized area is at most $1 + \rho$, it gives squared derivative error at most ζ in tile units. For the remaining tiles, the sum of their converted tile measures is bounded by ρ^{-1} times the positive excess. The crude disk estimate bounds their energy by a constant times area plus the bounded boundary energy. Thus their total converted energy is at most

$$C_M(1 + \rho^{-1}) (\text{total converted positive excess}) + o(1).$$

Make the positive excess small after fixing ρ . This proves (4.22) also in the critical rank-two case. The finite-piece and edge regularity needed by the disk estimate will be checked below before the volume gluing.

Rank two below the critical exponent. Suppose $k = 2$ and $1 \leq p < 2$. The macro selection has already fixed finitely many plane sections. On each S , first discard points where the tangential derivative differs substantially from the injective y -columns of D_i . On the retained set its two singular values lie between bounds depending only on M . The trace test and Chebyshev's inequality show that the discarded integral of $1 + |D_y f|^p$ is at most $C_M \tau^p h^2$. Select compact C^1 Lusin jet sets in the interiors of the original tiles, losing an arbitrarily small additional integral. The jets extend to ambient charts b by adding a normal column with the ambient orientation of f . Their derivative and inverse derivative bounds are C_M . Thus the leftover part of the section satisfies

$$h^{-2} \int_{\text{leftover}} (1 + |D_y f|^p) \leq C_M \tau^p + \text{an arbitrarily small error.} \quad (4.26)$$

Choose disjoint compact neighborhoods on the source and target sides. The new compacts avoid the original tile edges. They also avoid P_0 of every shell edge: when those images lie in S , their y -coordinates belong to the tile boundaries. Consequently all these fixed graph images have positive clearance from the new compact neighborhoods.

Inside each section use a new macro grid of side s and a fine grid of side ls , with $s \rightarrow 0$ only after the macro data have been fixed. To respect the original tile seams, insert their fixed coordinate levels, omit shifted s -levels within $s/4$ of a seam, and subdivide each remaining gap into N equal pieces. The resulting rectangles have controlled aspect ratios. Near a seam, every nonfixed subdivision level has a shift coefficient of magnitude at least $1/N$, so its translation density is bounded for fixed l . Only boundedly many such levels occur per seam in an $O(s)$ band. Therefore their edge trace integrals, multiplied by ls , are bounded in expectation by C_l times the section derivative integral on that shrinking band. The fixed seam lines have finite trace integrals and their converted weights tend to zero. Both contributions vanish as $s \rightarrow 0$.

Apply Lemma 4.3 with $m = 2$ on the new compacts, simultaneously with the preparation of H_0 . All target radial and vertex motions avoid the already-budgeted fixed graph because of the clearance just established. On each successful micro block, make the radial source change only as a parametrization of the section, and keep its controlled core. Parametrize the remaining planar edges at constant speed, except for the prescribed core edges, and apply Lemma 3.4 to the remaining two-dimensional cells. It applies because $p < 2$. Do the same on fallback micro cells. These changes fit together to form a tile-preserving bi-Lipschitz map σ_Y of the section. They are not inserted as an additional ambient precomposition.

On the compact jets in the micro cores, the map equals b , apart from the bounded-derivative caps of Lemma 4.3; its derivative is $D_y f$ there. The cap fraction and shell contributions are $O_M(l)$ in section units. The derivative is bounded by C_M on the remaining core portions. The shell spokes have image length $O_M(\gamma s)$, so constant-speed parametrization gives their required bounded energy. Ordinary outer and fallback edges use the individual high-probability length bounds, followed by the planar collapse estimate. The latter has the form

$$\int_D |Dg|^p \leq C(ls) \int_{\partial D} |D_t g|^p + \text{a chosen error}.$$

Translation averaging and (4.26) therefore give

$$h^{-2} \int_Y |Dh_Y - D_i|_y^p \leq C_M l + C_M \tau^p + o_s(1) + \text{an arbitrarily small error.} \quad (4.27)$$

Here failed micro blocks contribute vanishing cost by their failed area and bounded compact jets; elsewhere the ordinary converted trace cost is bounded by the leftover integral. The constants in that latter bound depend on M , not on later micro-chart widths or their number. The seam errors were handled separately above. Multiplying by h and summing proves (4.22). On the original tile edges, the same construction gives an upper p -energy bound by a fixed constant times the original trace p -energy plus chosen errors. There are no micro output motions on those edges.

4.5 From section cores to volume cells

We now keep all finite section data and output motions fixed. On a rank- k macro core use

$$G_\lambda(y, z) = V(P_\lambda(\sigma_Y(y), z)), \quad (y, z) \in B. \quad (4.28)$$

Since P_0 is flat on B , its argument here is

$$(\sigma_Y(y), z_0 + \lambda(z - z_0)).$$

Lemma 4.2 gives

$$DG_\lambda \longrightarrow [Dh_Y, 0] \quad \text{in } L^p(B) \quad (\lambda \downarrow 0). \quad (4.29)$$

The outer map V has finitely many closed-piece C^1 formulas on the compact configurations: this is true for H_0 and for each radial, angular, and cap map. The inner maps are uniformly Lipschitz for fixed σ_Y , and their derivatives have the stated pointwise limit. This verifies the hypotheses even if the limiting map has rank one or two.

The core boundary values preserve the images of its individual facets. Indeed, set

$$w_\lambda = V \circ P_\lambda, \quad T_B(y, z) = (\sigma_Y(y), z).$$

Then $G_\lambda = w_\lambda \circ T_B$ on B . Since σ_Y preserves the tile complex, T_B preserves each fine facet and edge of ∂B . In particular,

$$G_\lambda(F) = w_\lambda(F) \quad \text{for every such facet } F.$$

Thus these boundary values can be prescribed on the adjacent shell cells without changing their base face images. The product formula need not be extended through the shell; its base map there is $w_\lambda = V \circ P_\lambda$. Apply Lemma 4.2 separately on its edges and faces, with limiting parameter map P_0 . Their lengths and areas converge to the transferred budgets, or to smaller upper bounds when rank is lost.

We record the boundary scaling of the prescribed core data. Facets normal to a kernel coordinate have limiting energy bounded by

$$Ch^{2-k} \int_Y |Dh_Y|^p. \quad (4.30)$$

For $k = 2$, facets normal to a y -coordinate use the original tile-edge parametrizations on ∂Y ; their total energy per macro block is $O_M(h^2)$. For $k = 1$ their limiting energy is zero. The total limiting edge energy on the subdivided ∂B is

$$O_M(h/l). \quad (4.31)$$

For the critical rank-two case this follows from the uniform edge speed bounds. Below the critical exponent use instead the individual original tile-edge energy estimates just proved; the trace tests bound their sum. Edges replicated in kernel directions have the same tangential data, and kernel-axis edges have zero limiting energy. Multiplying (4.30) by lh and (4.31) by $(lh)^2$ gives $O_M(l)$ times the macro-volume contributions, using (4.22) for the first expression. Small positive λ preserves these estimates with arbitrarily small error.

Lemma 4.6 (Completion of uncontrolled cells). *Let D be an uncontrolled source three-cell of scale H in the construction. Its base embedding $v : D \rightarrow \mathbb{R}^3$ is bi-Lipschitz and is given by finitely many C^1 formulas extending to neighborhoods of their closed pieces. Write g for the prescribed or subsequently chosen boundary data. On every prescribed core face F , assume that $g|_F = v|_F \circ s_F$, where $s_F : F \rightarrow F$ is a bi-Lipschitz self-homeomorphism preserving each edge. These self-maps agree on shared edges. When $p = 2$, their edge restrictions have derivatives with essential limits almost everywhere. Then the cell can be reparametrized bi-Lipschitzly, preserving its base image and all prescribed boundary data, with derivative cost at most a constant times*

$$\begin{aligned} & H \sum_{F \text{ prescribed}} \int_F |D_t g|^p + H^2 \sum_{e \subset \partial D} \int_e |D_t g|^p \\ & + \mathbf{1}_{\{p=2\}} H \sum_{F \text{ uncontrolled}} \text{Area}(v|_F) \end{aligned} \quad (4.32)$$

plus any prescribed positive error. An unprescribed edge of length comparable to H may be given energy at most $C \text{length}(v(e))^p H^{1-p}$. The constants depend only on p and the fixed shape bounds, and adjacent cells can be completed with identical shared-face data.

Proof. First reparametrize every unprescribed edge at constant speed, keeping common vertices. On an unprescribed face, cone-extend these edge changes. If $p < 2$, Lemma 3.4 in dimension two gives face energy bounded by CH times its boundary p -energy plus a chosen error. If $p = 2$, use the crude estimate of Lemma 3.3; after scaling it bounds the face energy by a constant times its base image area plus H times its edge squared energy. Shape-controlled polygons are converted to square parameters by fixed bi-Lipschitz closed-piece smooth charts. Prescribed faces are left with their given parametrizations. Make one choice per shared face.

The disk hypotheses follow from the actual base and edge regularity. Restricting finitely many closed-piece C^1 formulas gives the almost-everywhere essential limits on a base face and the tangential limits at its parameter edges. On each fixed edge subdivision, normalized arclength has derivative with essential limits almost everywhere. Bi-Lipschitzness bounds that derivative away from zero, and the inverse arclength map has the same property.

For prescribed kernel edges, the source self-map is the restriction of T_B . In the rank-one case it is the interval arclength change; in the critical rank-two case it is the arclength change on each fixed J -subinterval of the section edge. The disk parametrizations fix the tile boundaries, so they leave these edge changes intact. Kernel-axis edges use the identity. Reusing these source self-maps

at positive λ retains their essential derivative limits, even though the resulting curves need not have constant speed for the base map w_λ . On radial-core facets the prescribed map already equals v , so $s_F = \text{id}$. Thus all prescribed edges needed for a critical face satisfy the stated condition. Coning the edge changes preserves the compatibility required by Lemma 3.3.

The assembled boundary map g gives the face-preserving bi-Lipschitz self-homeomorphism $v^{-1} \circ g$ of ∂D . Apply Lemma 3.4 to this boundary self-map in dimension three, where $p \leq 2 < 3$. It contributes a factor H times the true boundary-face energy. Substituting the preceding face estimates gives (4.32). This last collapse requires the radial compatibility of the base parametrization and a fixed bi-Lipschitz boundary map; it does not require stronger smoothness of the entire boundary reparametrization. Allocate the intermediate errors so their converted sum is below the prescribed cell error. $\square$

4.6 Global choices, summability, and strong convergence

Proof of Theorem 4.1. We first construct locally bi-Lipschitz approximants; smoothing is performed only after their global derivative costs have been estimated. If necessary, compose with a reflection to fix an orientation convention, and undo it at the end.

Order of the fixed data. Let $\varepsilon_0 > 0$ be a tail tolerance, to be sent to zero. Choose the compact rank truncation and its bound M in Lemma 4.4. Choose l so small that all the $O_M(l)$ contributions in volume and section shells and in cap regions are below a prescribed auxiliary tolerance. Recall that $\gamma = l/K_0$.

With M, l fixed, choose the sharp derivative tolerances for the line and plane tiles. The speed bounds entering those tolerances are fixed in terms of M . When $k = p = 2$, choose the further edge subdivision J large enough for the desired boundary closeness. Choose η small enough for the sharp length and area coefficients, then τ small enough for all trace errors, including the factors depending on l, η, J . Next choose δ in (4.13) small enough, refine the compact pieces, and choose their nested excess neighborhoods so that (4.14) is small after multiplication by every now fixed inflation factor. These bounds do not depend on the widths of the neighborhoods or on the number of their compact pieces. Only then choose h small and make the macro success selections. The planar micro configurations, when present, are chosen afterward, using (4.26) and then small s . Their ambient chart constants are controlled by M .

Fix the bounded radial area weights and compact clearance margins before choosing the target budget mesh. Choose that mesh using Proposition 3.15 and Lemmas 3.16, 3.17, and 3.18. This determines H_0 , with the required exact affine germs. The small radial parameters, angular attenuation parameters, and vertex contractions are then chosen from the actual finite configuration. In kernel blocks choose λ after the section parametrizations and all output maps have been fixed. Finally complete the uncontrolled cells using Lemma 4.6, with summable positive error prescriptions. Lemma 1.2 permits the locally finite fine choices and these summable prescriptions.

Controlling ordinary and failed cells. We give the accounting that makes this order of choices sufficient. For ordinary cells meeting no compact jet set and no enlarged special neighborhood, source sampling bounds the converted trace integrals in expectation by

$$C_p \int_{\Omega \setminus \bigcup_i K_i} |Df|^p. \quad (4.33)$$

On the simultaneous individual curve-bound event, Hölder's inequality bounds the converted edge costs by these source trace integrals. When $p = 2$, the conditional expectation over the later target

mesh bounds the converted face-area budgets by the corresponding source traces. Substituting in (4.32) gives precisely the codimension conversion factors H for faces and H^2 for edges. The constants in (4.33) are ordinary mesh and budget constants, independent of M, l, η .

In a failed macro block, the compact-jet integrands are bounded by a constant depending on M . For the ordinary fine grid, the total converted size of edges and faces in that block is $O(h^3)$, with fixed factors from any special target normalization allowed. The converted source trace integrals over its compact-jet portions are therefore bounded deterministically by Ch^3 . After the source mesh and failed blocks have been selected, the target area-budget expectation applies to these fixed traces. On the individual curve-bound event, Hölder's inequality gives the same bound for converted edge costs. Thus the small failed-block volume controls their total without requiring a source trace law conditional on failure. On off-compact portions, discard the failure indicator before applying the unconditional source trace bounds.

Whenever a nonuniversal constant can occur, its whole source trace lies in one of the chosen enlarged source neighborhoods for small enough meshes. Indeed its image meets a buffered anisotropic or radial-motion zone, whose inverse image is compactly inside that neighborhood; the small cell diameter and the fine initial-map errors preserve this inclusion. Thus even an edge length-power estimate can pay its finite factor on the *whole* edge's Hölder integral, then split this integral into the bounded compact-jet part and the small excess part. By (4.14), all these excess contributions are as small as prescribed. No later large constant multiplies the unprotected tail in (4.33).

For full-rank radial blocks, Lemma 4.3 gives the small wedge budget and the contracted spoke lengths. On compact points of a failed full-rank block, $F = \text{id}$, so their image points cannot belong to another block's inset angular support. Outer ordinary traces of retained blocks have the $O_M(l)$ converted cost already computed. Any off-compact surface portion affected by an angular map is estimated by the uncapped envelope, integrating in u_j before the later clearance-dependent caps. Its conditional mean is bounded for the fixed trace template. This gives the same small excess estimate. Nonincident edges avoid the angular supports and hence never pay a power of that angular envelope. Target-mesh suprema of the fixed bounded weights are allowed an arbitrarily small enlargement error by a still finer target mesh.

The kernel shells are controlled by Lemma 4.5, (4.30), and (4.31). Their prescribed face and edge contributions in (4.32) are $O_M(l)$ in bulk units. The radial volume shells have the same bound, since their prescribed core faces have bounded derivatives and their radial edges have length $O_M(\gamma h)$. All estimates are relative to the original straight product or frustum cell sizes; no shape bound is imposed on the intermediary image cells.

Simultaneous choices. Each estimate just used bounds a sum of nonnegative errors. In the sharp line and plane cases, this sum is the positive excess, not the signed total excess alone; the lower bounds in the section argument are what allow that replacement. Choose the expectations smaller than the desired tolerances with enough slack, and use Markov's inequality for the finitely many aggregate objectives. The source and micro configurations are fixed first, using the trace, failed-volume, and angular-weight bounds. The target sample is chosen next, meeting the expectation objectives together with the arbitrarily high-probability individual curve bounds. Countably many ordinary curves are handled with summable failure allowances and local mesh upper bounds. Generic slicing and intersection conditions have full measure on their templates; the star conditions have already been accounted for by the small failed volume.

In particular, the sharp disk tolerances are selected using fixed speed bounds. Make the positive excess small only after their area threshold ρ is chosen. The converted measure of bad sharp tiles and their crude energy are then small by the explicit ρ^{-1} estimate above. This avoids any assertion

that the lattice intersection count itself converges pointwise on a wrinkled trace. These choices yield deterministic configurations with all the stated bounds simultaneously.

The resulting homeomorphism. Let V be H_0 after the ambient radial-angular expansions and the vertex contractions. Their supports are disjoint within each family. Distinct compact configurations have separated target neighborhoods; the much later micro-section supports were chosen smaller still, and avoid all fixed graph data. In the rank-three volume blocks precompose by the radial source squeezes. In kernel blocks use P_λ as the shell base map and (4.28) on the core. Section micro squeezes enter only through σ_Y . Complete all remaining edges, shared faces, and volume cells as in Lemma 4.6.

Each change is a self-parametrization relative to the same base image, with matching boundary values. The meshes are locally finite, and each interior compact meets only finitely many pieces. Finite local straight cell geometry and the finite local bi-Lipschitz constants give a locally bi-Lipschitz map across the interfaces. It is a homeomorphism of Ω onto the same target as V . The initial H_0 is onto Ω' , and every output change is a self-homeomorphism supported inside Ω' . Consequently the assembled map H is onto exactly Ω' .

Derivative error. Let $\mathcal{C}$ be the union of the controlled volume cores. The preceding bounds imply

$$\int_{\Omega \setminus \mathcal{C}} |DH|^p \leq C_p \varepsilon_0 + o(1), \quad (4.34)$$

where the auxiliary errors, including $O_M(l)$, can be made as small as prescribed in their stated order. The original derivative has the same small bound on this complement: outside the compacts use (4.12), while on the compacts the shell fraction and failed volume are small and the jets are bounded.

On a full-rank core, $H = b_i$ except on the small bounded-derivative caps. On its compact jet set, $Db_i = Df$; on the remaining portion the chart derivative is bounded and the excess integral is small. Thus the derivative difference there is small. On a kernel core, (4.29) and (4.22) give closeness to D_i with the correct fiber factor h^{3-k} . Equation (4.13) compares D_i with Df on K_i , and the excess-neighborhood bound treats the remaining points. Choose the finitely many λ 's sufficiently small after the other data. Combining these estimates with (4.34) yields

$$\int_{\Omega} |DH - Df|^p \leq C_p \varepsilon_0 + o(1). \quad (4.35)$$

All uncontrolled-cell errors were chosen summably. The displayed bounds therefore also establish $\int_{\Omega} |DH|^p < \infty$, rather than only local Sobolev regularity.

Value convergence. The source self-parametrizations move points only within cells or blocks whose maximum diameter tends to zero. The initial budget map can have arbitrarily small value error, and the target motions are supported in boxes of arbitrarily small diameter in physical coordinates. On every interior compact, continuity of f then gives value convergence. Since the domains are bounded and all images lie in the fixed bounded target, dominated convergence gives L^p convergence on Ω . Choose the meshes and value prescriptions so its error and (4.35) sum to less than the prescribed ε ; this proves (4.1). □

5 Completion on the fixed target

Proof of Theorem 1.1. Fix $1 \leq p \leq 2$ and $\varepsilon > 0$. Apply Theorem 4.1 with sufficiently small error, obtaining a locally bi-Lipschitz homeomorphism $H : \Omega \rightarrow \Lambda$ in $W^{1,p}$. Apply Theorem 2.1 to H with a smaller error. The triangle inequality gives a smooth diffeomorphism $g : \Omega \rightarrow \Lambda$ such that

$$\int_{\Omega} (|g - f|^p + |Dg - Df|^p) dx < \varepsilon.$$

Both approximation theorems give global $W^{1,p}$ membership and retain the target for each approximant. In particular, the proper-degree argument in Lemma 2.4, used during smoothing, establishes bijectivity before any limit is taken; the nonsingular smooth differential then supplies the smooth inverse. Apply the construction with $\varepsilon = 2^{-j}$ to obtain (1.1). $\square$

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